IE University
School of Science & Technology

The Demographic Tide: Retirement Flows and Price Impact in an Inelastic Equity Market

Antonio de Ipiña Sanchez
Bachelor of Data and Business Analytics

Supervised by:
Prof. Manuele Leonelli

Executive Summary

The consensus that retiring baby boomers would not depress equity markets had formed before passive index investing captured roughly a third of U.S. equities (Chinco & Sammon, 2024). It also predates the rise of auto-enrolment and target-date funds, which built a mechanical pipeline between demographics and equity markets. This study asks when aggregate retirement account outflows will overtake inflows into passive equity index funds, in an inelastic market where one dollar of flow moves prices by an estimated five dollars (Gabaix & Koijen, 2023).

This study builds a five-module accounting model calibrated from demographic projections, participant-level retirement plan data, industry fund flow statistics, and equity issuance data. Price impact is then estimated through the Gabaix-Koijen inelastic markets framework. Across 25,000 Monte Carlo paths, the median demographic net flow trajectory crosses zero between 2040 and 2041. The annual price drag reaches −0.11% of market capitalisation by 2050, less than one-hundredth of the long-run equity return. Including corporate buybacks, total net flows remain positive throughout the horizon in 95.5% of simulations. But the buyback offset is procyclical and contracts in downturns.

No meltdown materialises under the model's conservative TDF-based assumptions. But the transition is real. The total DC system is already in net outflow, and under observed retiree equity allocations, the sensitivity test shows the crossover advancing to 2025. The Sobol variance decomposition reveals that the voluntary withdrawal multiplier, not market return, is the decisive structural parameter. This points toward retirement withdrawal behaviour, not market forecasting, as the key uncertainty. These results matter for TDF providers, policymakers, and institutional investors alike.

Table of Contents

Executive Summary **2**

1. Introduction **5**

1.1 The Problem 5

1.2 The Gap and Research Question 6

1.3 Preview and Roadmap 7

2. Literature Review **8**

2.1 The Demographic Asset Meltdown Debate 8

2.2 The Inelastic Markets Revolution 9

2.3 Retirement Savings and Withdrawal Behaviour 11

2.4 The Institutional Channel 12

2.5 The Gap 13

3. Data **13**

3.1 Data Sources and Calibration 15

3.2 Data Limitations 17

4. Methodology **18**

4.1 Model Architecture 18

4.2 The Five-Module Pipeline 19

4.3 Monte Carlo Simulation Design 23

4.4 Passive Share Forecasting: ARIMA 25

4.5 Regression Evidence 26

4.6 Global Sensitivity Analysis: Sobol Indices 26

4.7 Methodological Limitations 27

5. Results **28**

5.1 Flow Trajectories and Crossover Timing 29

5.2 Buyback Offset 33

5.3 Price Impact 33

5.4 Scenario and Stress Analysis 34

5.5 Variance Decomposition 37

5.6 Regression Evidence 39

6. Discussion **42**

6.1 The Central Finding 42

6.2 Robustness and Sensitivity 43

6.3 Implications 45

6.4 Scope, Limitations, and Future Work 46
7. Conclusion 48
References 51
Appendices 53
Appendix A: Data Sources and Parameter Tables 53
Appendix Figures 56

1. Introduction

Seventy-three million baby boomers, followed by early Generation X, are entering retirement. The financial system they pass through was built for inflows, not outflows. In the early 2000s, economists debated whether this population wave could depress equity markets. The general consensus was no. That consensus, though, was formed before two developments changed the picture. First, passive index investing now accounts for roughly a third of all U.S. equity holdings, parking assets in vehicles that do not respond to price signals. Second, automatic enrolment and target-date funds have built a mechanical pipeline between demographics and equity markets that simply did not exist when the original debate took place. In a market where demand is far more inelastic than previously recognised, these mechanical flows move prices by multiples of their dollar value.

1.1 The Problem

The baby boom has long been described as a pig moving through a python, a massive population bulge straining every institution it passes through. Today it reaches the retirement system. The baby boomers (born 1946 to 1964) and early Generation X (born 1965 to 1970) are the first large cohort in American history to build their retirement savings through employer-sponsored defined-contribution (DC) plans. Between 2025 and 2050 the population aged 65 and older will grow from 63 million to 82 million, a 30% increase, while the working-age population (ages 25 to 64) will grow from 173 million to 184 million, roughly 7% (U.S. Census Bureau, 2023) (Figure 1). At the same time, the passive share of U.S. equities has risen from close to zero in 1990 to approximately 33.5% (Chinco & Sammon, 2024), driven by lower fees, the Pension

Protection Act of 2006, and the growing evidence that most active managers underperform benchmarks.

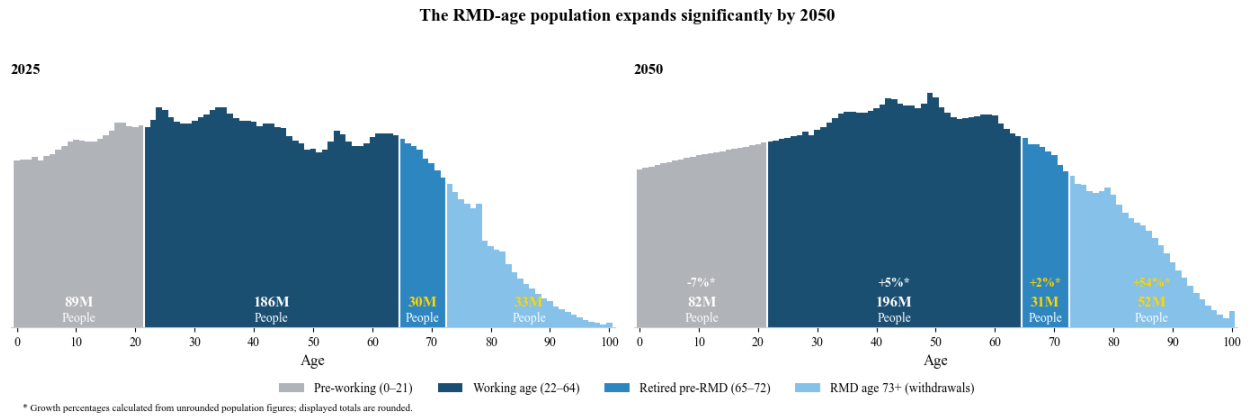

Figure 1: Working-age versus retirement-age population trends, 2025-2050.

These two trends converge in the retirement system. Contributions flow in through target-date fund defaults. Withdrawals flow out through mandatory distributions. Both are channelled through passive, price-insensitive vehicles. The system functions as long as inflows exceed outflows. But as the 65+ cohort expands, outflows will approach or exceed inflows. The shift is not only demographic. It is structural. The institutional plumbing connecting retirement savings to equity markets ensures that outflows pass through the same price-insensitive channels that carried the inflows.

1.2 The Gap and Research Question

Three bodies of research bear on this, but they have not been connected into a single forward-looking projection model. These are the demographic meltdown literature (Abel, 2001; Geanakoplos et al., 2004; Poterba, 2001), the inelastic markets literature (Chinco & Sammon, 2024; Gabaix & Koijen, 2023; Haddad et al., 2025), and the retirement behaviour literature (Madrian & Shea, 2001; Parker et al., 2023; Poterba et al., 2011). The tools to connect them are

recent: the Gabaix-Koijen multiplier was circulated in 2021, Haddad et al.'s elasticity measurements in 2025, and the SECURE 2.0 mandates were legislated only in 2022.

The question this study asks is whether and when aggregate retirement account outflows overtake inflows into passive equity index funds, and what the net flow trajectory looks like from 2025 to 2050. The model treats the crossover as a testable outcome, projecting probability distributions across 25,000 simulation paths. The analysis focuses on U.S. domestic DC plan flows into passive equity vehicles, the most mechanised and predictable institutional channel. The total DC system already disburses more than it receives. Net cash flow from DC plans into equity mutual funds has been negative every year since 2015 (ICI, 2025). The TDF-mediated channel modelled here remains positive under glide-path assumptions that understate current retiree equity holdings. The sensitivity test in Section 5.4 quantifies this gap. This study projects when that crossover becomes permanent, even under these conservative assumptions.

1.3 Preview and Roadmap

To answer this, the study constructs a five-module accounting model that traces U.S. domestic DC plan flows into passive equity vehicles, from contributions through to price impact, propagating uncertainty across 25,000 simulation paths. Section 2 reviews the demographic, inelastic markets, and retirement behaviour literatures, identifying the gap this study fills. Section 3 describes the data sources and their limitations. Section 4 develops the five-module methodology and the Monte Carlo, ARIMA, regression, and Sobol analyses. Section 5 reports results. Section 6 discusses implications, robustness, and limitations. Section 7 concludes.

2. Literature Review

Three literatures converge on whether demographic flows can move equity prices. The idea was proposed and largely dismissed in the early 2000s. What justifies revisiting it is that both market structure and the measurement tools have changed since then.

2.1 The Demographic Asset Meltdown Debate

The asset meltdown hypothesis proposes that as boomers retire and sell their holdings, the resulting sales depress equity prices. The main academic response appeared in a 2001 special issue of the Review of Economics and Statistics. It was largely sceptical. Poterba (2001) combined age-specific wealth holdings from the Survey of Consumer Finances with Census population projections to construct a projected asset demand variable. His core finding is a null result. Population age structure does not significantly predict asset returns, though he does find a more substantial relationship for price levels. Poterba's null doesn't undermine this analysis. His population ratios are too broad to capture the institutional channels through which retirement generates equity selling and his sample predates the passive revolution entirely. His own conclusion calls for research into these specific channels.

A common objection at the time was simpler. Retirees do not actually sell all their holdings. Many hold assets until death and pass them to heirs, so the shares stay in the market and no net sales pressure emerges. In the same issue, Abel (2001) shows this reasoning is incomplete. In an overlapping-generations model, when a retiree sells shares, the sale depresses prices, but a younger buyer provides offsetting demand. Net impact is zero. When a retiree bequeaths shares instead, the heir receives what they would otherwise have bought, both the sales pressure and the buying demand disappear together. The two paths lead to the same

equilibrium price. Bequests do not prevent demographic effects on valuations. They simply rearrange who holds the shares.

Geanakoplos et al. (2004) provided the most comprehensive theoretical framework, introducing the MY (middle-aged to young) ratio as a predictor of equity valuations. Their "favoured cohort effect" shows that large cohorts face unfavourable terms of trade at every stage. When a large cohort enters the labour market, its collective demand for equities bids prices up; when the same cohort retires and sells, the supply surge depresses them. Brooks (2004) argues that the MY ratio's correlation with prices could reflect coincident structural changes rather than demographics, that is a valid criticism, but it does not affect the present analysis, which models specific institutional channels rather than relying on the MY ratio.

This early literature established the theoretical mechanism but lacked the empirical tools to test it. Those tools arrived with the inelastic markets revolution.

2.2 The Inelastic Markets Revolution

Gabaix and Koijen (2023) argue that the U.S. stock market is inelastic, estimating that one dollar of net flow increases aggregate market capitalisation by approximately three to seven dollars, with a central estimate of five. They identify this using Granular Instrumental Variables (GIV) across 13 sectors. The multiplier arises because a large share of equity is held by investors who rarely adjust in response to price changes. These include passive index funds, pension funds following policy portfolios, foreign central banks, and inattentive households. Their general equilibrium model matches the observed equity premium, return predictability, and excess volatility, which lends theoretical credibility to the empirical estimate. The GIV identification strategy is not without criticism. It relies on sector-level flow variation that may not perfectly

separate mechanical flows from those carrying fundamental information, and the resulting multiplier spans a wide confidence interval ($M \approx 3$ to 7). Two further caveats matter. The multiplier was estimated from an inflow-dominated period, so its applicability to outflows is untested. And the hypothesis itself remains debated.

Sabbatucci et al. (2025) provide complementary evidence using BrightScope data on individual employer retirement plans, estimating a flow multiplier of 12 to 13 with feedback effects. This suggests $M = 5$ is conservative. Their analysis operates at the stock level using cross-sectional variation in plan-level passive allocation, whereas this study projects aggregate flow trajectories over time. The questions are complementary but methodologically distinct.

Haddad et al. (2025) independently confirm low demand elasticity at the individual stock level and document that it declined by approximately 11% between 2001 and 2020 as passive investing grew. Their strategic response parameter $\chi \approx 3$ means that active investors partially offset the elasticity lost to passive growth, but about one-third passes through to equilibrium prices. Empirical validation through Russell index reconstitution events yields a pass-through estimate of 0.35, close to the theoretical 0.33 presented by Haddad et al.

Chinco and Sammon (2024) show that the true passive share is approximately 33.5%, roughly double the commonly reported figure. Standard measures miss closet indexers and institutional separate accounts. Mitchell et al. (2007) document that these price displacements persist for weeks to months, as arbitrage capital responds slowly to flow-driven mispricings. Together, these papers paint a consistent picture. Flows move prices with a large multiplier. That multiplier is growing. And the effects persist. Two independent approaches, Gabaix and Koijen's aggregate time-series and Haddad et al.'s cross-sectional analysis, converge on this conclusion

through distinct empirical strategies. That convergence strengthens the case that low elasticity is a robust feature of modern equity markets.

2.3 Retirement Savings and Withdrawal Behaviour

If markets are inelastic, the magnitude of the price impact depends on how mechanical the flows are. The retirement savings literature suggests DC plan flows are highly mechanical.

Madrian and Shea (2001) showed that automatic enrolment raised 401(k) participation from approximately 37% to 86% at a single large employer. The vast majority of these new participants retained the default contribution rate and fund allocation. Madrian and Shea attribute this to procrastination, status quo bias, and complexity-driven reliance on defaults. This behavioural inertia is what connects retirement savings to the inelastic markets framework. Participants rarely engage with their accounts regardless of market conditions.

That inertia now flows predominantly through target-date funds. TDFs have grown from $70 billion in 2005 to approximately $2.0 trillion in mutual fund assets alone by year-end 2024 (ICI, 2025), with additional holdings in collective investment trusts bringing the total TDF market to roughly $3.5 to $4.0 trillion. Parker et al. (2023) showed that this growth creates a mechanical channel between demographics and equity markets. TDFs allocate based on age, not prices. Giglio et al. (2021), using Vanguard data on a large panel of retail investors, confirmed that beliefs about expected returns vary substantially across investors but portfolio allocations respond weakly. Poterba et al. (2011) documented actual withdrawal behaviour using Health and Retirement Study microdata, showing that retirees decumulate more slowly than life-cycle models predict (Modigliani & Brumberg, 1954). These findings calibrate the voluntary withdrawal rates in the outflow module. Both Madrian and Shea and Poterba et al. draw on

single-employer or survey samples, which raises questions about generalisability. Aggregate ICI flow data can partially address this.

2.4 The Institutional Channel

The institutional machinery connecting these literatures has been built over the past decades through regulatory change. The shift from defined-benefit to defined-contribution pension plans transferred investment decisions from professional managers to millions of individual participants. DC plans now total $12.4 trillion compared with $3.3 trillion in private-sector DB plans at year-end 2024 (ICI, 2025), covering around 70 million active 401(k) participants. Auto-enrolment, mandated for new plans under the SECURE 2.0 Act of 2022, pushes participation to 94% compared with 64% under voluntary enrolment (Vanguard, 2025b).

These contributions flow overwhelmingly into target-date funds, which adjust their equity allocation from approximately 90% for young workers to 30% in the final retirement stage (total equity, including international allocations) (Vanguard, 2025b). TDFs do not yet dominate total accumulated DC assets; the remaining balance sits in self-directed portfolios established before auto-enrolment. But they overwhelmingly capture new contributions, which is the flow channel the model tracks. TDFs are also macrocontrarian, systematically rebalancing against market moves, buying when the market drops and selling when it rises. Parker et al. (2023) documented that roughly 45% of rebalancing occurs within the same month.

On the outflow side, the required minimum distribution (RMD) forces money out of these accounts. RMDs are mandatory annual withdrawals from tax-deferred accounts such as 401(k)s and IRAs, and they increase with age. When an RMD is taken from an account invested in an index fund, the custodian sells shares pro rata, mechanically transmitting demographic pressure

into equity selling. Together, these features ensure that the entire savings-to-withdrawal lifecycle operates without active participant decisions. The result is precisely the kind of price-insensitive flow that the inelastic markets hypothesis identifies as a mover of price levels.

2.5 The Gap

These three literatures, and the institutional channel that connects them, have developed largely in isolation. This study assembles them into a single projection framework using a partial-equilibrium accounting model rather than a general-equilibrium approach. An overlapping-generations model would endogenise prices and demographics simultaneously but would require aggregating away the cohort-level TDF mechanics that are central to the analysis. The accounting framework preserves this granularity at the cost of omitting equilibrium feedback, which Section 6.2 addresses directly. The model tracks dollars from working-age contributions through TDF allocations to retirement withdrawals. Net flows are translated into price impact through the Gabaix-Koijen flow multiplier, adjusted over time using the Haddad et al. pass-through as passive ownership grows. Output uncertainty is decomposed through multi-horizon Sobol variance analysis.

3. Data

Turning this pipeline into a quantitative projection requires data on each stage. All primary sources except Bloomberg and FactSet are publicly available. Full parameter tables, descriptive statistics, and all processing code are available in Appendix A and the project's GitHub repository (github.com/Aipinas/capstone-demographic-flows). Table 1 summarises the sources and the model components they calibrate.

Table 1

Data Sources and Calibration

Source	Module	Variables / Parameters
U.S. Census Bureau (2023)	1	Population by single-year age, 2025–2060; main, high, low, zero immigration scenarios
Vanguard, How America Saves (2025b)	2	Age-specific participation rates, total contribution rates (9.3–14.6%), TDF glide paths (12 vintages), auto-enrolment adoption
BLS National Compensation Survey (2024)	2	DC plan access rate (72%); age-specific employment-to-population ratios
BLS Employment Cost Index (2025)	2	Nominal wage growth rate (2.84%/yr CAGR, 2001–2025); 65+ employment trend (+0.18 pp/yr)
Morningstar TDF Landscape (2025)	2	Cross-provider validation of Vanguard glide-path representativeness
IRS Publication 590-B, Table III (2024)	3	RMD divisors by age (deterministic withdrawal floor)
Federal Reserve SCF (2022)	3	Retirement account balances and ownership rates by age bracket
Vanguard, How America Retires (2025a)	3	Voluntary withdrawal rates by age group; retiree equity allocations (52% at ages 65–69)
Federal Reserve Z.1, Table F.103 (2025)	4	Quarterly net equity issuance, 1996:Q4–2025:Q3 (buyback yield: 0.98% ± 0.72%)
S&P Dow Jones Buyback Reports (2024)	4	Gross repurchase data (cross-validation); recession cut calibration (30%/50%)
Bloomberg Business Intelligence (2025)	5	Semi-annual passive equity AUM, 56 asset-class series, 2012–2025
FactSet, FT Wilshire 5000 (2025)	5	Historical U.S. equity market capitalisation (passive share denominator)
Chinco & Sammon (2024)	5	Corrected passive ownership share (33.5%); scaling factor 1.583
Damodaran (2025)	MC	Annual U.S. equity returns 1928–2024 (arithmetic mean 11.8%, std dev 19.5%)
ICI Fact Book, Table 20 (2025)	Regression	Quarterly net new cash flow into U.S. equity mutual funds, 2007:Q3–2024:Q3
EBRI/ICI 401(k) Database (2022)	Validation	Actual equity allocations by age (57% at ages 60–69); balance cross-checks

Note. Module column indicates the primary model component calibrated from each source. MC = Monte Carlo simulation parameters. All primary sources except Bloomberg and FactSet are publicly available. Full parameter tables are reported in Appendix A.

3.1 Data Sources and Calibration

The demographic backbone of the simulation is the U.S. Census Bureau's 2023 National Population Projections, which provide population counts by single-year age for 2025 through 2060 in four scenarios. These are a main projection and three immigration variants (high, low, zero). The simulation uses the main projection as the central case. Demographic projections over a 25-year horizon are among the most reliable inputs in the model. The majority of the relevant population (those who will be of working or retirement age between 2025 and 2050) has already been born. Immigration is the primary source of demographic uncertainty. The three variant scenarios allow the model to bracket it (Module 1).

Inflow parameters come primarily from Vanguard's "How America Saves" (2025b), which covers approximately 1,400 plans and nearly 5 million participants. The simulation extracts age-specific participation rates, total contribution rates including employer matching (approximately 9% to 15% of salary depending on age), auto-enrolment adoption rates, and TDF usage patterns. The equity allocation of inflows follows Vanguard's published TDF glide paths across 12 target-date vintages (2020 through 2070 plus Income). The U.S. equity share ranges from 53.6% for the youngest workers (Target Retirement 2070) to 18.5% for the Income fund (Figure 2). Morningstar's TDF Landscape report provides cross-provider validation, confirming that Vanguard's glide path is representative of the industry. Plan access rates come from the National Compensation Survey (Bureau of Labor Statistics, 2024), which reports that 72% of private-sector workers have access to employer-sponsored retirement benefits, the vast majority through DC plans (Module 2).

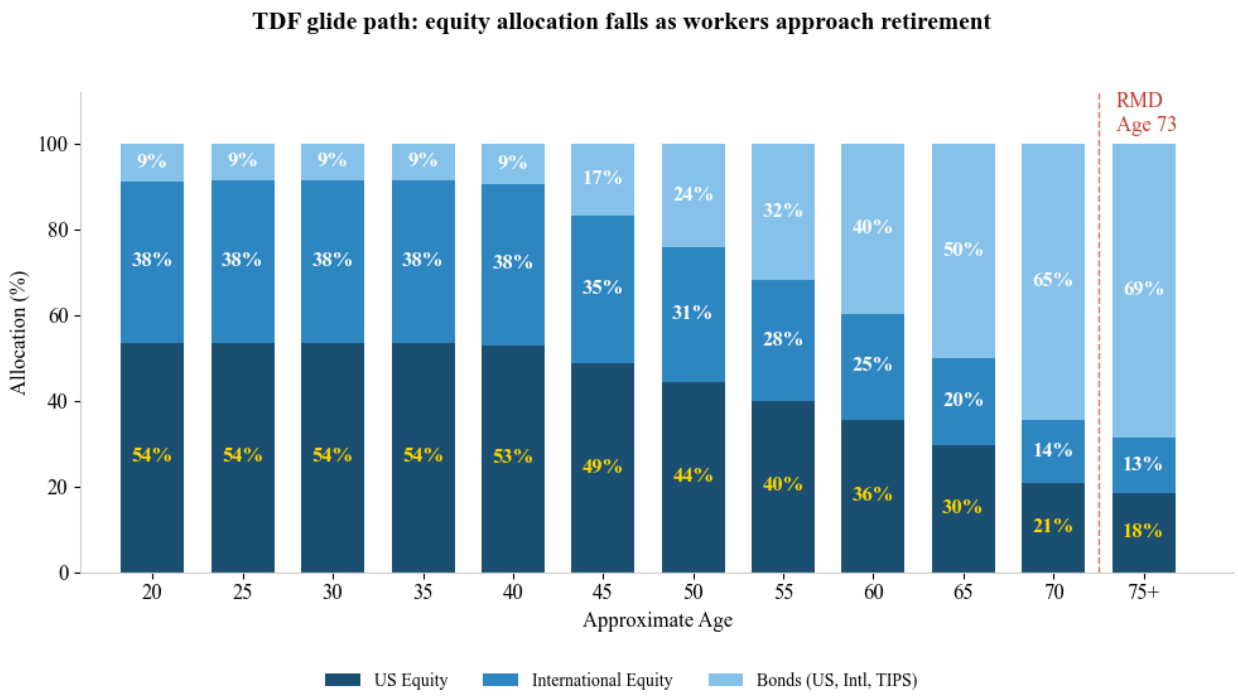

Figure 2: TDF glide path. U.S. equity allocation by target retirement date, from 53.6% (2070 vintage) to 18.5% (Income fund).

Outflow parameters combine three sources. The IRS Publication 590-B Uniform Lifetime Table (Table III) provides deterministic RMD divisors by age. The Federal Reserve's 2022 Survey of Consumer Finances (SCF) provides retirement account balances by age bracket, used to calibrate the initial balance distribution for outflow calculations. Vanguard's "How America Retires" (2025a) provides empirical data on actual withdrawal behaviour beyond RMDs. These data calibrate the base voluntary withdrawal rates by age group. In the Monte Carlo simulation, these base rates are scaled by a multiplier drawn from U(0.5, 1.5), producing effective voluntary withdrawal rates that range from half to one-and-a-half times the calibrated baseline (Module 3).

The buyback offset in Module 4 is calibrated from the Federal Reserve's Enhanced Financial Accounts (Z.1 release, Table F.103), which reports quarterly net equity issuance and

retirement from 1996Q4 to 2025Q3. The resulting net buyback yield is approximately 0.98% with a standard deviation of 0.72%. S&P Dow Jones Quarterly Buyback Reports provide gross repurchase data for cross-validation. FactSet provides historical U.S. equity market capitalisation data (FT Wilshire 5000) used as the denominator for the passive share calculation (Module 4 & 5).

The time-varying multiplier in Module 5 depends on the trajectory of passive equity ownership. Bloomberg's Business Intelligence database provides semi-annual passive equity assets under management (AUM) across 56 asset-class series from 2012 to 2025. This series feeds the ARIMA passive share forecast. Chinco and Sammon's (2024) corrected passive share estimate of 33.5% as of 2021 anchors the calibration; the simulation's 2025 starting point of approximately 35% reflects continued growth in the Bloomberg series, with a scaling factor of 1.583 (the ratio of corrected to raw industry figures) applied to convert standard measures to the true passive share (Module 5).

3.2 Data Limitations

The Vanguard data represents a single plan administrator. This skews the participant base towards larger employers and higher balances compared with the full universe of DC plans. ICI industry-wide data confirm that 68% of 401(k) participants hold TDFs (ICI, 2025), supporting the model's assumption that TDFs are the dominant allocation vehicle. The model applies TDF glide-path equity shares to all cohorts. This correctly describes the allocation for participants auto-enrolled after the Pension Protection Act of 2006, but it understates the equity exposure of current retirees who established their portfolios before TDFs became the default. EBRI/ICI data show that actual total equity allocations exceed the TDF assumption at every age, with the gap

widest for retirees: 57% actual versus 21% in the model for participants in their sixties (Vanguard, 2025b). As TDF-era cohorts age into retirement, this gap will close. The TDF sensitivity test in Section 5.4 quantifies the near-term effect by applying observed retiree equity allocations to the 2025 to 2045 window. ICI does not report industry-wide contribution rates by age, so the Vanguard-calibrated rates (9.3% to 14.6%) cannot be directly validated against an independent source.

The Federal Reserve's Survey of Consumer Finances reports retirement account balances at the household level, not the individual level. This may overstate per-person holdings in dual-earner households.

ICI Table 20 (used in the regression analysis) reports mutual fund equity outflows only, excluding ETF redemptions. The regression coefficient on population aged 65+ should be read as a lower bound on the demographic sensitivity of total equity outflows.

4. Methodology

The model translates the data described in Section 3 into a forward-looking projection through five sequential modules, one for each stage of the retirement flow pipeline.

4.1 Model Architecture

An accounting model was chosen because the research question requires tracking cohort-level flows year by year through the institutional pipeline. Every assumption must be visible.

The five modules form a sequential pipeline. Demographic projections (Module 1) feed inflow estimation (Module 2) and outflow estimation (Module 3), which aggregate into net flows with a buyback offset (Module 4) and translate into price impact through the Gabaix-Koijen

multiplier with the Haddad et al. time-varying elasticity framework (Module 5). The simulation tracks 76 individual age cohorts (ages 25 through 100), ageing one year per step.

The analysis runs across four Jupyter notebooks and three Python modules (simulation_engine.py, model_runner.py, sobol_runner.py). A shared simulation engine ensures parameter consistency across the Monte Carlo and Sobol analyses. Run order and dependency structure are documented in the repository README.

4.2 The Five-Module Pipeline

4.2.1 Module 1: Demographic Projections. Module 1 draws on the Census projections described in Section 3.1, aggregating single-year ages into six working-age brackets (under 25, 25-34, 35-44, 45-54, 55-64, 65+) for Module 2 and seven retirement-age brackets (55-59, 60-64, 65-72, 73-74, 75-84, 85-94, 95+) for Module 3. The simulation uses the mid scenario as the central projection and treats low and high scenarios as bounds in the 3x3 scenario matrix. The mid scenario receives 40% weight as the Census Bureau's central projection, with high and low immigration each at 30%. Appendix Table A7 shows that the crossover probability has a spread of 2.4 percentage points, ranging from 89.1% to 91.5% across scenarios, driven by the working-age population that determines inflow growth.

4.2.2 Module 2: Inflow Estimation. For each simulation year and age group, the inflow into U.S. equities through the retirement system is computed as a contribution funnel. The funnel is expressed as:

$$Inflow(g,t) = Population(g,t) * Employment(g) * PlanAcces * Participation(g) * DeferralRate(g) * Salary(g) * TDF_{EquityShare}(g)$$

Population is the cohort size from Module 1, Employment is the age-specific employment ratio, PlanAccess is the fraction of workers with access to a DC plan, Participation is the enrolment rate, DeferralRate is the total contribution rate including employer matching, Salary is median earnings by age, and $TDF_{EquityShare}$ is the U.S. equity allocation corresponding to the cohort's expected retirement year. All parameters vary by age and are calibrated as described in Section 3. The resulting 2025 baseline is $265 billion, consistent with the approximately 37% equity share of the $716 billion in total DC plan contributions reported for 2023 (ICI, 2025). The simulation tracks U.S. equity flows only. International equity allocations within TDFs are excluded because the study focuses on U.S. market impact.

Inflows grow through nominal wage growth of 2.84% per year, the mean annual increase in the Employment Cost Index from 2001 to 2025 (Bureau of Labor Statistics, 2025). The employment ratio for workers aged 65 and older increases by 0.18 percentage points per year (capped at 25%), reflecting observed trends in Bureau of Labor Statistics employment data (2008 to 2024).

4.2.3 Module 3: Outflow Estimation. Retirement outflows arise from three channels. Required minimum distributions (RMDs), voluntary withdrawals in excess of RMDs, and the systematic de-risking of TDF allocations as participants age through the glide path. The outflow is expressed as:

$$Outflow(g,t) = Population(g,t) * Ownership(g,t) * Balance(g,t) * \max\left(RMD_{Rate}(g), VoluntaryRate(g)\right) * TDF_{EquityShare}(g)$$

Population is the cohort size from Module 1 and Ownership is the fraction of each age group that holds a retirement account. Balance is the mean account balance, which grows or shrinks with market returns. RMD_{Rate} and VoluntaryRate are the mandatory and voluntary

withdrawal rates, and TDF_EquityShare is the U.S. equity allocation at the appropriate point on the glide path. The max() operator means whichever rate is higher applies. All parameters are age-specific and calibrated as described in Section 3. Voluntary withdrawal rates are set at 1% of balance for ages 55-59, 2% for 60-64, and 3% for 65-72. Above age 73 the RMD rate exceeds any voluntary rate and becomes the binding constraint. In the Monte Carlo simulation, voluntary rates are scaled by a multiplier drawn from U(0.5, 1.5) to reflect uncertainty in how quickly retirees actually draw down their accounts. The SECURE 2.0 Act raised the RMD starting age from 73 to 75 for individuals born in 1960 or later. This creates a two-year window of reduced mandatory outflows for the tail end of the boomer cohort. The simulation incorporates this discontinuity, but the effect is modest. By the late 2030s, outflows are dominated by retirees aged 80 and older whose withdrawal rates exceed the 73-year-old threshold.

Ownership rates among retirees rise over time as pension-era cohorts give way to the 401(k) generation. Older cohorts converge toward younger cohorts' ownership levels at 2.7% per year, capturing this generational turnover.

Account balances are updated each simulation year by applying the stochastic market return. In strong years, account balances grow and subsequent RMDs are mechanically larger. In crash years, balances shrink and RMDs decline, partially self-correcting the outflow. The 2025 baseline aggregate outflow is $188 billion.

4.2.4 Module 4: Net Flow Aggregation and Buyback Offset. The demographic net flow in each simulation year is the difference between aggregate inflows and outflows, summed across all age cohorts. The model then incorporates corporate stock repurchases as a partially offsetting force. Buybacks are treated as a yield on market capitalisation so that buyback volume scales with market size. The net buyback yield of approximately 0.98% is calibrated directly from Federal Reserve Z.1 net equity issuance data (2020-2024), which already accounts for dilution from new share issuance and stock-based compensation. The total net equity flow is the sum of the demographic net flow and the buyback contribution. This determines whether the market experiences net buying or net selling in each simulated year. The buyback offset is modelled as procyclical. When annual returns fall below −10%, the effective buyback yield is reduced by 30%, and below −20% by 50%, calibrated to observed corporate behaviour during the 2008-2009 and 2020 downturns (S&P Dow Jones Indices, 2024). The buyback buffer disappears in exactly the market conditions where an offsetting force is most needed.

4.2.5 Module 5: Price Impact Estimation. The flows modelled in Modules 2 and 3 are mechanical and do not respond to price signals. The inelastic markets hypothesis holds that such flows move aggregate market capitalisation by a multiplier M = 1/ζ, where ζ is the price elasticity of demand. The model draws the base multiplier uniformly from [3, 7], spanning the approximate 95% confidence interval around the central estimate of M = 5 (Gabaix & Koijen, 2023). A stress test at M = 2 provides the lower bound.

As passive ownership grows, more of the lost elasticity passes through to prices. The effective elasticity in each simulation year is:

$$\zeta_{eff} = \zeta_{base} * (1 - PassThrough * \Delta_{passive})$$

Where $PassThrough = 1/(1 + \chi * Active\ share)$, ActiveShare is one minus the passive share, $\chi = 3$ is the strategic response parameter from Haddad et al. (2025), and $\Delta_{passive}$ is the change in passive share relative to the 2025 baseline of 35%. The effective multiplier $M_{eff} = 1/\zeta_{eff}$ is floored at $\zeta = 0.05$ to prevent numerical instability. At 2025, around one-third of lost elasticity passes through. By 2050 this rises to one-half. Under the central scenario ($M_{base} = 5$, passive share rising from 35% to 67%), the effective multiplier grows from 5.0 to approximately 6.0 by 2050.

The preceding five modules define the flow pipeline. The remaining subsections describe how uncertainty is propagated.

4.3 Monte Carlo Simulation Design

The model applies a Monte Carlo simulation because the research question asks when and whether, not what is the expected value. The answer is a probability distribution over crossover years, not a point estimate. The simulation generates 25,000 independent paths over the modelled horizon from 2025 to 2050.

Three structural parameters are drawn once per path and held constant. These are a demographic scenario (mid, high, or low immigration, weighted 40/30/30), a base multiplier M from U(3, 7), and a passive share growth term from N(0, 1) applied to the ARIMA confidence band. Four parameters vary independently each simulation year. Equity returns are drawn from N(11.8%, 19.5%), calibrated from annual U.S. equity returns over 1928 to 2024 (Damodaran, 2025), buyback yields from a truncated normal centred on 0.98%, voluntary withdrawal multipliers from U(0.5, 1.5), and inflow shocks from N(1.0, 0.07). The return distribution uses the arithmetic mean (11.8%) because compounding with volatility erodes median growth. A

portfolio earning 11.8% per year on average with 19.5% volatility compounds to the same median terminal wealth as a riskless 9.9% return. Deterministic scenarios therefore apply 9.9% directly. A fixed random seed ensures exact reproducibility, and the crossover probability stabilises by approximately 10,000 paths (Appendix Figure A-1).

When projected net DC equity flows are negative, a regime absent from the historical sample, the model applies the Gabaix-Koijen multiplier as a return drag. The adjustment is $r_{adj} = r_t + M * \min(Net_{flow}, 0)/mktcap$. Positive flows produce no adjustment. Their price effect is already embedded in the historical return distribution. The feedback is self-dampening. Negative flows reduce returns, which reduce balance growth, which in turn reduce future outflows.

Table 2

Monte Carlo Parameter Distributions

Parameter	Distribution	Calibration
Demographic scenario	Categorical	Mid (40%), High (30%), Low (30%) immigration
Equity return	N(0.118, 0.195)	Arithmetic mean and std dev, Damodaran (2025), 1928–2024
Buyback yield	N(0.0098, 0.0072)	Fed Z.1 Table F.103, 2020–2024; truncated [0.1%, 3.5%]
Inflow shock	N(1.0, 0.07)	Multiplicative; ±2σ covers 0.86–1.14
Voluntary withdrawal multiplier	U(0.5, 1.5)	Scales base voluntary rates from Vanguard (2025a)
Base multiplier M	U(3, 7)	95% CI of Gabaix & Koijen (2023) central estimate
Passive share z-score	N(0, 1)	Applied to ARIMA forecast confidence band

Seed = 42 for reproducibility. Returns are drawn from a normal distribution; observed equity returns exhibit fatter tails, which would widen confidence intervals without changing the central estimate.

4.4 Passive Share Forecasting: ARIMA

The model forecasts the passive ownership share of U.S. equities using an ARIMA model fitted to 27 semi-annual observations of passive fund AUM as a percentage of total U.S. equity market capitalisation. The data come from Bloomberg. The entire series is scaled by a factor of 1.583 to align with Chinco and Sammon's (2024) corrected estimate of 33.5%, accounting for closet indexing and quasi-passive strategies. This yields an adjusted passive share of approximately 35% in 2025 (Figure 3).

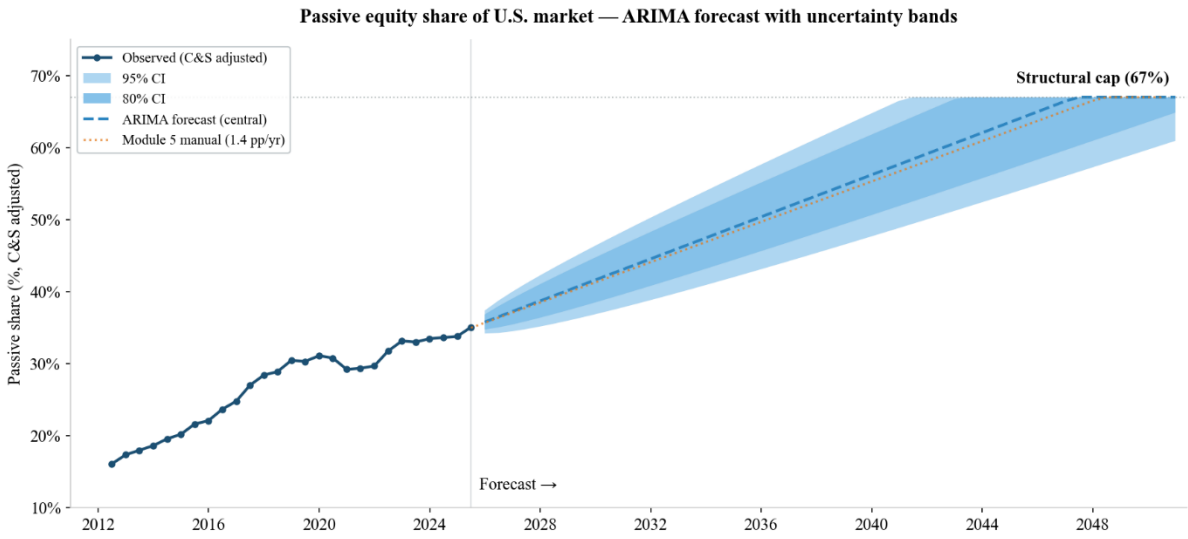

Figure 3: Historical passive share with ARIMA forecast, confidence intervals, and 67% cap.

Augmented Dickey-Fuller tests confirm non-stationarity in levels ($t = -1.65$, $p = 0.46$) and stationarity in first differences ($t = -3.71$, $p = 0.004$), motivating first-differencing. AIC selects ARIMA(0,1,0) with drift, the simplest specification. Passive share grows at approximately 0.46 percentage points per half-year plus noise, with no additional memory structure improving the forecast. Ljung-Box tests on the residuals show no autocorrelation at lags 4, 8, and 12 (all $p > 0.18$), and the one-step-ahead root mean square error is 1.93 percentage points. Appendix Table A5 reports the full model comparison. A 67% upper cap is imposed

because the ARIMA would otherwise overshoot. The cap represents approximately double the current U.S. passive share (Chinco & Sammon, 2024) and is imposed as a modelling assumption. No empirical evidence pins the ceiling precisely, but double the current level is a generous upper bound. Beyond that point, the remaining active base is assumed thin enough that profit opportunities would slow further adoption. Appendix Figure A-2 compares the 67% cap with an 80% alternative. The Sobol analysis confirms that passive share growth has a near-zero impact on net flow timing but contributes more meaningfully to price impact uncertainty. The forecast yields 50.4% by 2035 and 67.0% by 2050, at which point the cap binds. Each Monte Carlo path samples within the ARIMA confidence band to capture uncertainty in the pace of adoption.

4.5 Regression Evidence

The simulation projects forward. To validate the mechanism it models, a reduced-form OLS regression tests whether the ageing population has historically been associated with equity mutual fund outflows. The dependent variable is quarterly net new cash flow into U.S. equity mutual funds (ICI Table 20, 2007:Q3 to 2024:Q3, 69 observations). The control variable is the contemporaneous S&P 500 quarterly return. Newey-West standard errors with four lags address serial correlation. Full results are reported in Section 5.6.

4.6 Global Sensitivity Analysis: Sobol Indices

The Monte Carlo answers when and whether. The three literatures reviewed in Section 2 make competing claims about what drives the timing. The demographic work emphasises population structure and returns (Poterba, 2001; Geanakoplos et al., 2004), the retirement behaviour literature emphasises withdrawal patterns (Poterba et al., 2011), and the inelastic markets framework emphasises the pass-through parameter (Haddad et al., 2025). Sobol variance

decomposition adjudicates between these mechanisms by isolating each parameter's contribution to output variance. The conditional analysis, which fixes the return, is specifically designed to test whether the voluntary withdrawal multiplier dominates crossover timing once market noise is removed. The analysis uses Saltelli sampling (N = 2,048) in four configurations. An overall analysis varies all seven parameters simultaneously, with equity returns represented as a single draw applied to all 26 years. A 32-parameter analysis disaggregates returns into 26 individual annual draws, testing whether aggregating the return sequence into a single parameter understates its contribution through path interactions. A 57-parameter analysis further disaggregates voluntary withdrawal multipliers into 26 annual draws alongside the 26 annual returns, testing whether the dominance of returns is an artefact of returns being the only parameter that varies per year. A conditional analysis fixes the annual return at 9.9% and varies only the six structural parameters, isolating the drivers that matter once market noise is removed. The conditional Sobol uses the geometric mean (9.9%) rather than the arithmetic mean (11.8%) because a fixed return applied identically each year must match the compound growth rate of a typical stochastic path, not its expected single-year value. All four analyses are evaluated at three horizons (2030, 2040, 2050).

4.7 Methodological Limitations

Several equilibrium channels remain outside the model. The DC return feedback adjusts returns when net flows turn negative, but does not apply to positive flows even when they fall below historical norms. This modestly overstates balance growth in the transition years before crossover. The historical return distribution reflects a period of exclusively positive DC flows, and decomposing the DC channel's contribution to those returns remains a causal identification problem addressed in Section 6.4. Returns are drawn independently each year from a normal

distribution, which understates tail risk relative to observed equity return distributions. This omits the volatility clustering that would concentrate extreme returns and widen the crossover year distribution. Buyback yields, withdrawal multipliers, inflow shocks, and returns likely move together, particularly during recessions when all four deteriorate simultaneously. But the model draws them independently, likely understating tail risk. The Haddad $\chi = 3$ parameter captures active investor response in the cross-section, but whether it remains stable under sustained aggregate outflows is untested.

On the outflow side, the model applies TDF glide-path equity allocations to all retirees. This understates near-term outflows from self-directed participants, as discussed in Section 3.2. The model also excludes the SECURE Act's 10-year rule for inherited accounts, IRA rollovers, and household debt-driven early withdrawals, all of which would add to outflow pressure.

On the inflow side, contributions are modelled as acyclical, likely understating the procyclicality of net flows during downturns. Nominal wage growth is held constant at 2.84% per year, and the 2.7% annual ownership convergence rate for older cohorts is derived from a short observation window from 2013 to 2022 (SCF).

5. Results

The simulation answers the research question in three stages. When does the demographic crossover occur? Does it survive corporate buybacks? And what is the resulting price impact? Sections 5.1 through 5.3 address these in sequence, while 5.4 through 5.6 test robustness, decompose variance, and validate the mechanism against historical data.

5.1 Flow Trajectories and Crossover Timing

Table 3 reports the 2025 baseline.

Table 3

2025 Baseline Calibration

Metric	Value ($B)
Median inflow (Module 2)	265.2
Median outflow (Module 3)	188.3
Median demographic net flow	77.2
Median buyback offset (Module 4)	591.2

Note. Each column reports the independent median across 25,000 Monte Carlo paths. Net flow does not equal inflow minus outflow because the median of a difference is not the difference of medians. Buyback offset = net buyback yield × market capitalisation.

Both inflows and outflows grow over the projection horizon, but at different rates. Median inflows more than double from $265 billion to $579 billion by 2050, driven by nominal wage growth applied to salary-linked contributions. Median outflows nearly quadruple from $188 billion to $746 billion as the retirement-age population expands and account balances compound under positive returns.

The model's 2025 baseline positions the equity channel at a turning point, but the broader DC system has already crossed it. ICI data confirm this transition is underway at the aggregate level. Net cash flow from DC plans into equity mutual funds has been negative every year since 2015, reaching −$93 billion in 2019, −$241 billion in 2020, and −$228 billion in 2024, with no sign of reversal (ICI, 2025). The modelled TDF equity channel remains positive at +$77 billion in 2025, reflecting the conservative glide-path assumption. Under observed retiree equity allocations of approximately 50%, the sensitivity test in Section 5.4 produces immediate

crossover. The base case projects the structural regime the system is converging toward as auto-enrolled cohorts replace self-directed retirees. It is not a description of current conditions.

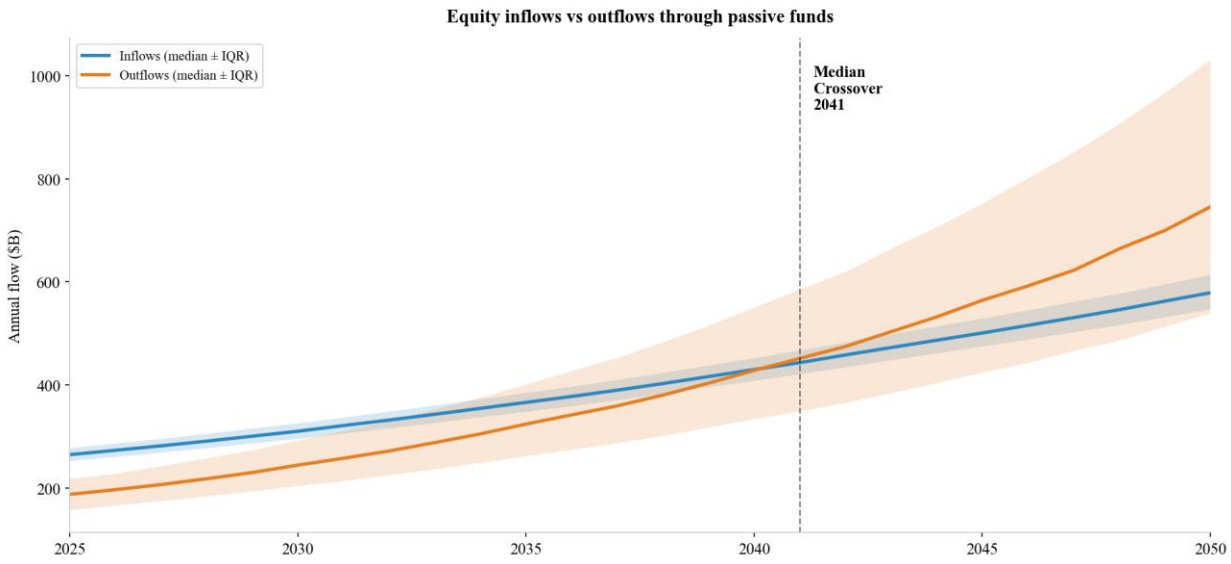

Figure 4: Median inflow versus outflow trajectories, 2025 to 2050.

The median net flow narrows from +$65 billion at 2030 to +$41 billion at 2035, to +$0.4 billion by 2040, and reaches −$167 billion by 2050. The median trajectory crosses zero between 2040 and 2041, consistent with the deterministic baseline (9.9% geometric mean, no volatility), which crosses between 2042 and 2043. The transition spans roughly a decade from the first signs of narrowing in the mid-2030s to sustained negative flows in the mid-2040s.

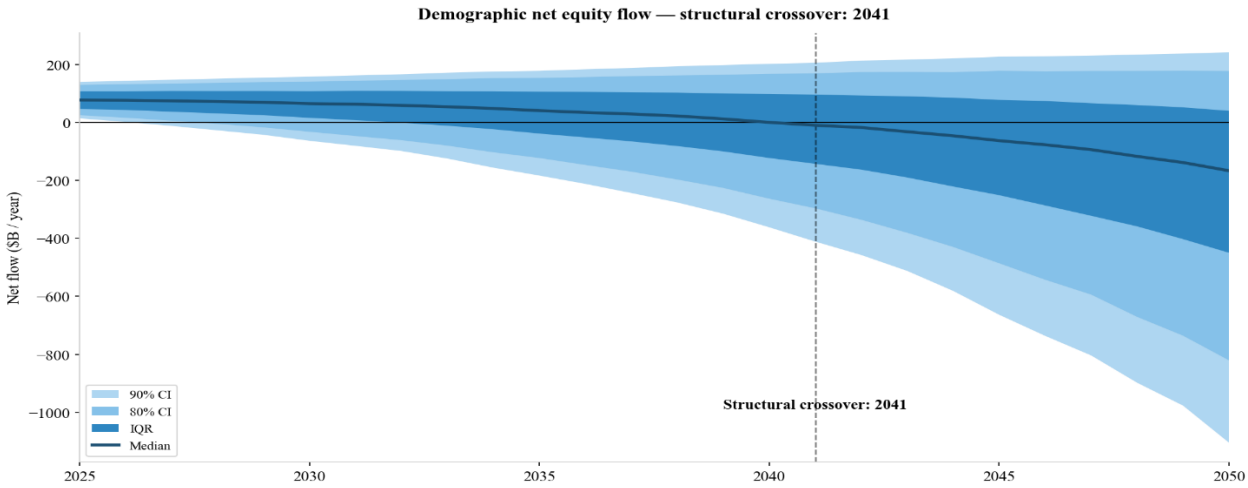

Figure 5: Fan chart of demographic net flows, 2025 to 2050, with percentile bands.

Return volatility causes negative years to appear well before the structural crossover. Of 25,000 simulations, 22,804 (91.2%) experience at least one year of negative demographic net flows before 2050, with a median first occurrence in 2031 (IQR: 2029 to 2036). These early crossovers are largely temporary. Below-average return years that inflate outflows relative to inflows before the structural trend reasserts itself. By 2050 the range is wide. The worst 5% of paths show over $1 trillion in annual net selling, while the best 5% still show positive net flows. That spread reflects 25 years of compounding return uncertainty.

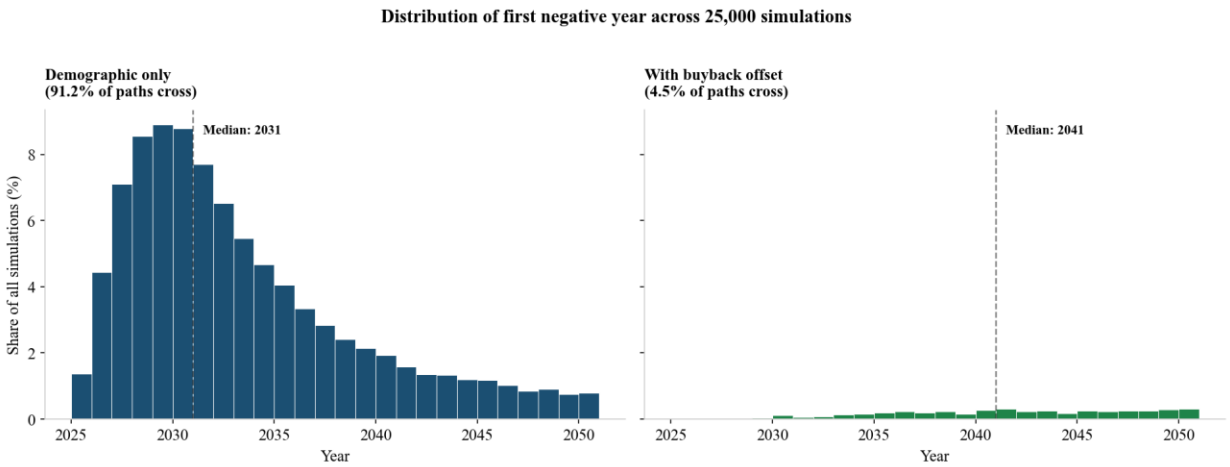

Figure 6: Distribution of crossover years across 25,000 simulations.

Appendix Table A7 disaggregates by immigration scenario. The probability of at least one negative year ranges from 89.1% (high immigration) to 91.5% (low immigration), a 2.4 percentage point spread driven by the working-age population that determines inflow growth.

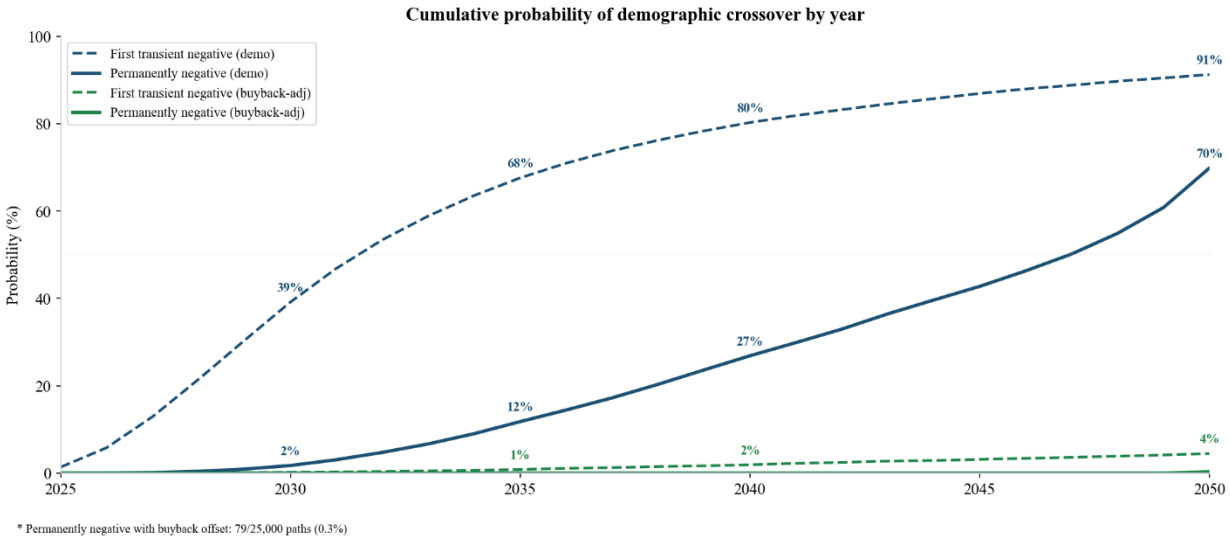

Figure 7: Cumulative crossover probability, demographic-only and buyback-adjusted.

5.2 Buyback Offset

When corporate buybacks are included as an offsetting flow, the demographic crossover largely disappears. Only 1,115 of 25,000 paths (4.5%) experience at least one year of negative total net flows by 2050, with a conditional median first occurrence of 2041. The buyback-adjusted probability is 0.2% at 2030, 1.9% at 2040, and 4.5% at 2050 (Figure 7). Of the 1,115 paths that cross, 1,036 (93%) are positive again by 2050. The crossover is temporary in the vast majority of cases. Only 79 paths (0.3% of all simulations) are in sustained deficit at the end of the horizon.

5.3 Price Impact

Demographic flows currently support equity prices. The median annual price impact is +0.57% of market capitalisation in 2025, declining to +0.29% at 2030 and +0.12% at 2035 as outflows grow. The median reaches essentially zero at 2040 (+0.001%) and turns negative by 2045 (−0.07%), reaching −0.11% by 2050. The IQR at 2050 spans −0.22% to +0.05% (Figure 8). Cumulatively, the demographic channel adds a median +$3.5 trillion in market capitalisation by 2040 before reversing to −$1.3 trillion by 2050 (IQR: −$15.1 trillion to +$9.1 trillion).

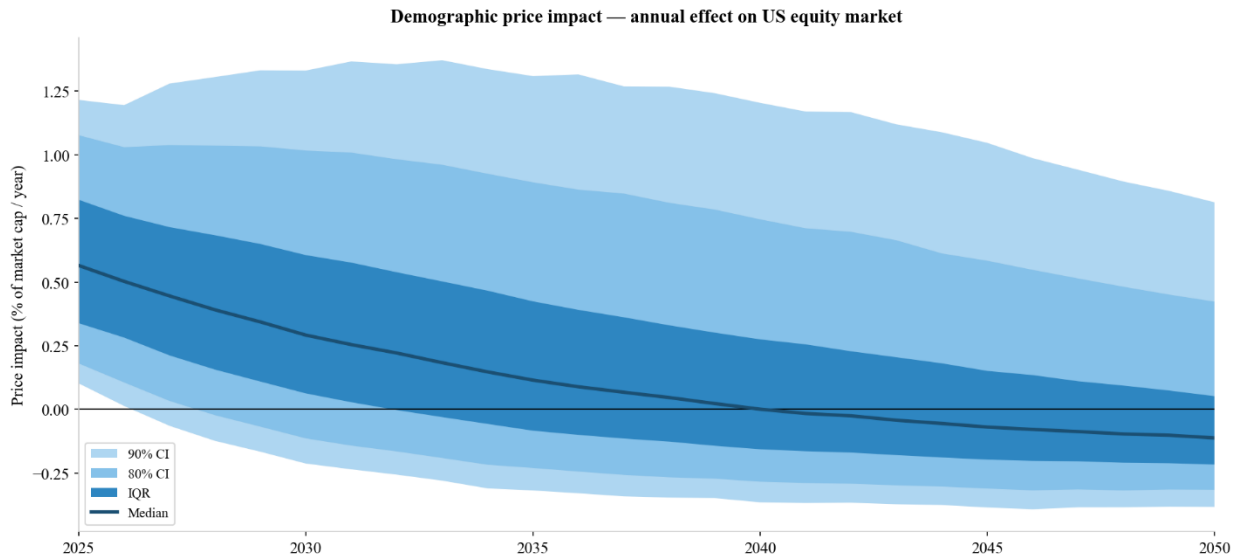

Figure 8: Price impact fan chart, 2025 to 2050 (% of market capitalisation per year).

5.4 Scenario and Stress Analysis

A 3×3 scenario matrix combines three base multiplier assumptions (M = 3, 5, 7) with three passive share growth rates (1.0, 1.4, 1.8 pp/yr), using mid-scenario demographics and the 9.9% geometric mean return. All nine scenarios produce a demographic crossover between 2042 and 2043, with net flows at 2035 uniformly positive at +$51 billion. The 2050 price impact ranges from −0.050% (M = 3, 1.0 pp/yr) to −0.122% (M = 7, 1.8 pp/yr). The 67% passive cap compresses the high-growth scenarios toward the base case (Table 4).

Table 4

3×3 Scenario Matrix: 2050 Price Impact (% of Market Capitalisation)

Scenario	Crossover	Net 2035	Net 2050	Impact 2050
M = 3, 1.0 pp/yr	2043	$51.0B	−$100.2B	−0.050%
M = 3, 1.4 pp/yr	2043	$51.0B	−$100.2B	−0.052%
M = 3, 1.8 pp/yr	2043	$51.0B	−$100.2B	−0.052%
M = 5, 1.0 pp/yr	2043	$51.0B	−$99.9B	−0.083%
M = 5, 1.4 pp/yr	2043	$51.0B	−$99.9B	−0.087%
M = 5, 1.8 pp/yr	2043	$51.0B	−$99.8B	−0.087%
M = 7, 1.0 pp/yr	2043	$51.0B	−$99.5B	−0.115%
M = 7, 1.4 pp/yr	2043	$51.0B	−$99.5B	−0.122%
M = 7, 1.8 pp/yr	2043	$51.0B	−$99.4B	−0.122%

Note. Deterministic scenarios using 9.9% geometric mean return, mid-scenario demographics. M = base multiplier (Gabaix & Koijen, 2023). Passive share growth in percentage points per year. The 67% passive cap compresses high-growth scenarios toward the base case.

A 30% market crash in 2030, 2035, or 2040 temporarily lifts net flows by reducing account balances and therefore RMD-driven outflows. The swing grows with the horizon, from +$40 billion for a 2030 crash to +$51 billion for 2035 and +$65 billion for 2040. This reflects the growing stock of retirement balances exposed to the crash.

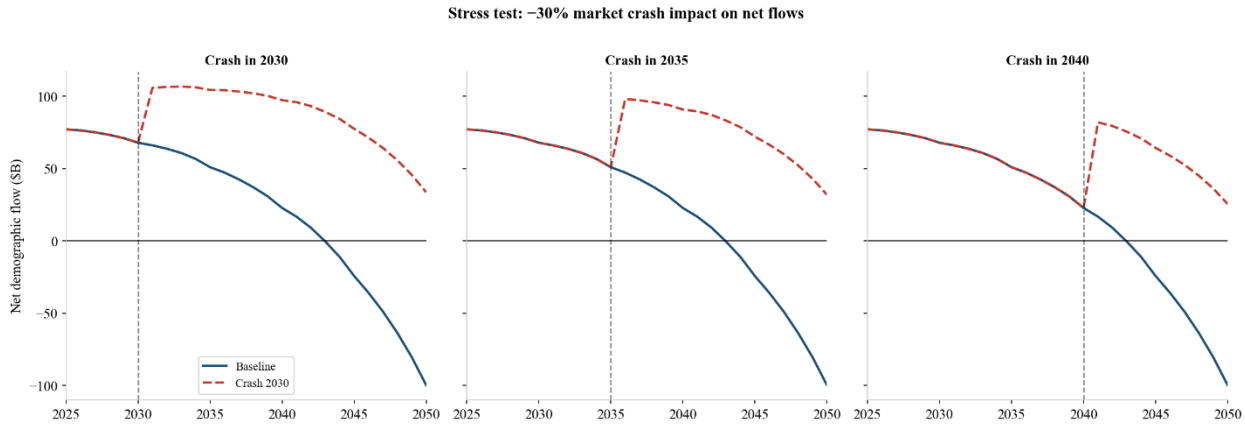

Figure 9: Crash stress test. Net demographic flow following a 30% crash in 2030, 2035, and 2040.

Recomputing price impact with a static M = 2 across all 25,000 paths tests whether the direction of the finding holds under a lower multiplier. Both M = 5 and M = 2 median

trajectories are essentially zero at 2040 and permanently negative thereafter, reaching −0.112% and −0.047% respectively by 2050. The ratio is approximately 2.4× across the full horizon. The multiplier governs the scale of the drag, not its direction.

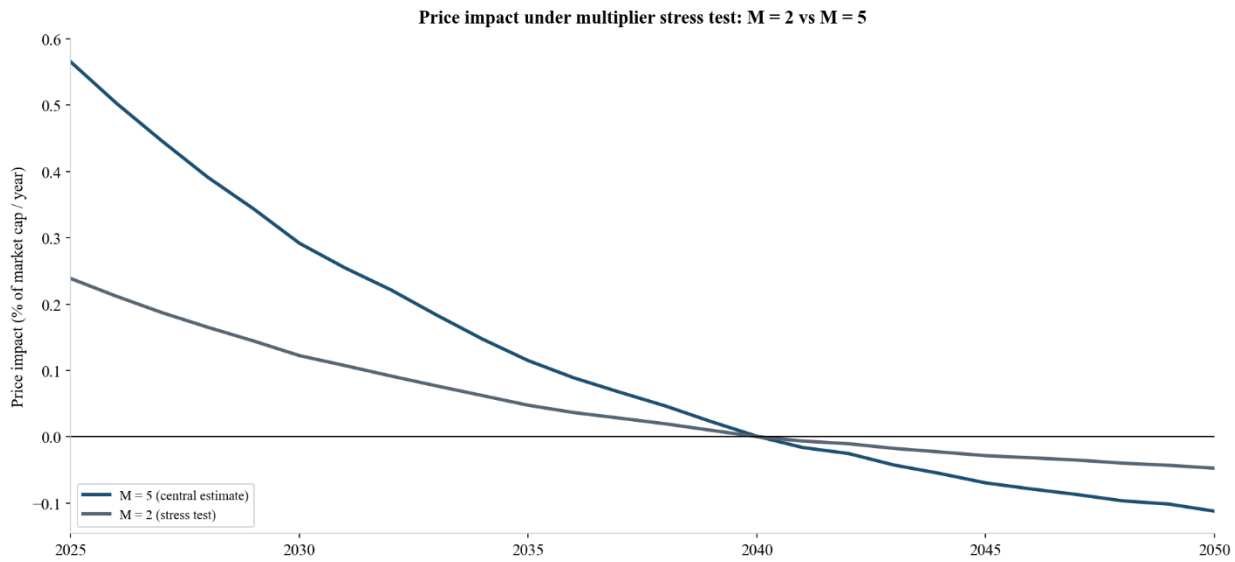

Figure 10: Multiplier symmetry stress test. Median price impact under M = 5 versus M = 2.

Two sensitivity tests apply observed retiree equity allocations with exponential decay over 2025 to 2045 to capture the gradual replacement of self-directed retirees by TDF-era cohorts. Under a 52% allocation, approximating observed Vanguard participant holdings of 52% at ages 65 to 69 and 45% at 70+ (Vanguard, 2025b), the crossover probability rises from 91.1% (in the 10,000-path subsample; 91.2% in the main run) to 100% and the median first negative year advances from 2031 to 2025. A conservative 36% alternative (double the TDF allocation) produces a crossover probability of 99.7%. The decay assumes TDF-era cohorts gradually replace self-directed retirees over a 20-year window. If self-directed allocations persist longer, the elevated outflows would extend further.

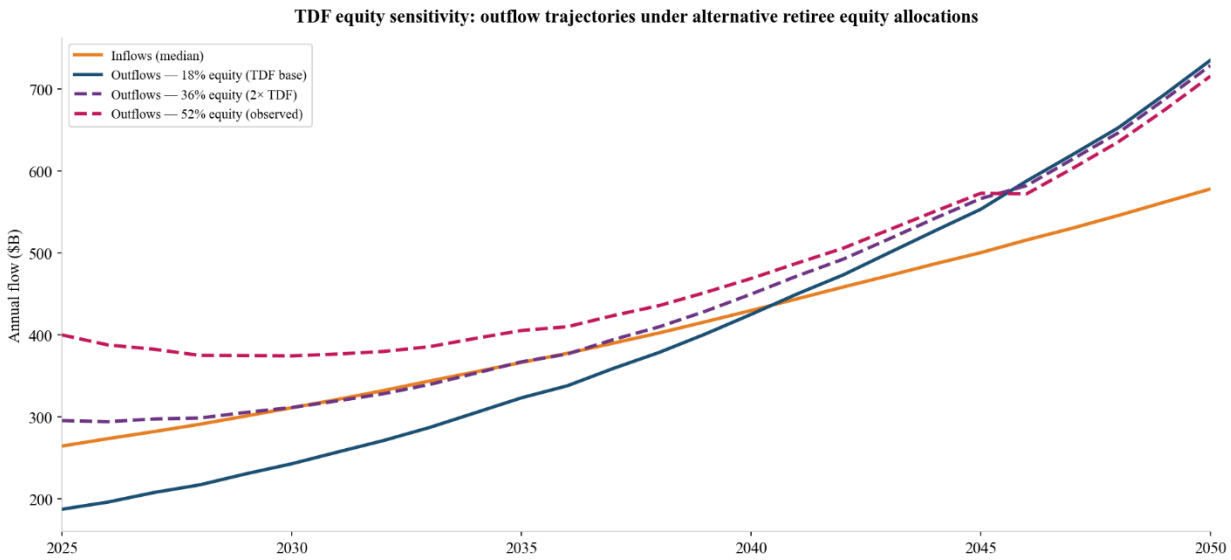

Figure 11: TDF equity sensitivity. Outflow trajectories under TDF baseline (18 to 21%), 36% (double TDF), and 52% (observed allocation), with exponential decay toward baseline over 2025 to 2045.

5.5 Variance Decomposition

The 57-parameter Sobol analysis varies both returns and voluntary withdrawal multipliers at annual resolution, matching the per-year draw structure of the Monte Carlo design. At the 2030 horizon, aggregate return and aggregate voluntary withdrawal contribute comparable shares of net flow variance (total-order Sobol index ST, measuring each parameter's share of output variance including interactions, at ST = 0.541 and 0.375 respectively), with inflow shock third at 0.115. By 2040, compounding return uncertainty dominates (aggregate return ST = 0.920, withdrawal ST = 0.165), and by 2050 return accounts for nearly all variance (ST = 1.064, with values exceeding 1.0 reflecting interactions between annual return draws). The voluntary withdrawal multiplier declines to ST = 0.099 by 2050 (Figure 12).

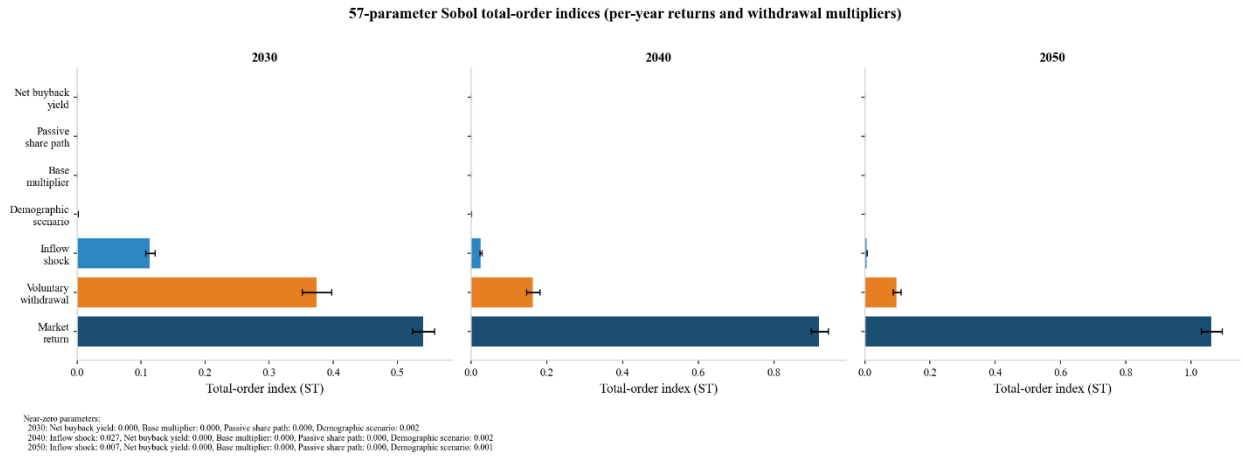

Figure 12: 57-parameter Sobol total-order indices at 2030, 2040, and 2050.

The convergence of the 7-parameter and 57-parameter analyses addresses a potential criticism. If aggregating the return sequence into a single annual draw had artificially inflated return dominance by capturing 26 years of compounding in one parameter, the 57-parameter analysis, which lets each year contribute independently, would have distributed variance more evenly across years. Instead, aggregate return ST is 0.971 in the 7-parameter analysis and 1.064 in the 57-parameter analysis at 2050, confirming that return dominance is a structural feature of the 25-year compounding horizon, not an artefact of parameter aggregation. Confidence intervals on all total-order indices are below ±0.05 across all four configurations, indicating adequate convergence at N = 2,048.

With returns fixed at the 9.9% geometric mean, the conditional analysis isolates the structural drivers. The withdrawal multiplier dominates at every horizon (S1 = 0.680 at 2030, 0.590 at 2040, 0.624 at 2050), with inflow shock accounting for most of the remainder (Figure 13).

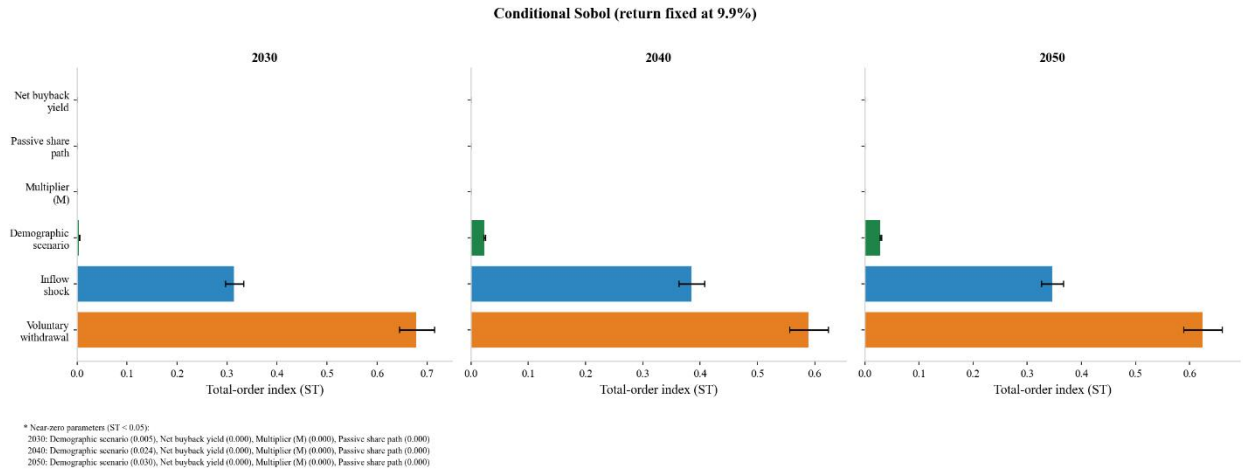

Figure 13: Conditional Sobol indices (return fixed at 9.9%) at 2030, 2040, and 2050.

For price impact, the base multiplier M contributes 6% to 11% of variance despite near-zero influence on net flows. Return dominates at all horizons (Figure 14).

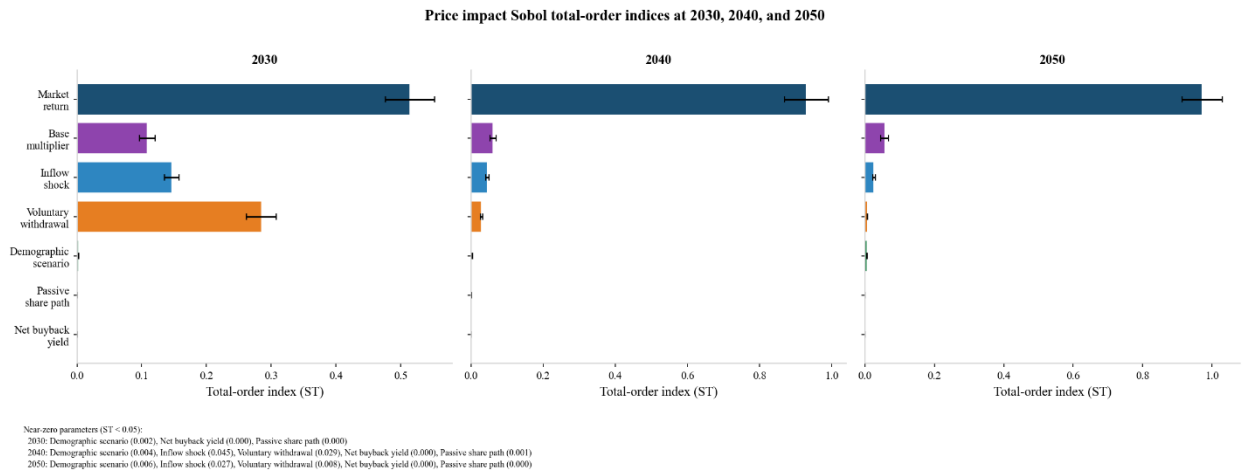

Figure 14: Sobol total-order indices for price impact at 2030, 2040, and 2050.

5.6 Regression Evidence

Independent of the simulation, the demographic channel should leave a trace in historical equity fund flows if the mechanism is real. An OLS regression of quarterly ICI Table 20 mutual fund equity net flows on population aged 65+ and S&P 500 returns (2007:Q3 to 2024:Q3, 69

observations) provides independent empirical validation (Table 5). Newey-West standard errors with four lags address serial correlation identified by Breusch-Godfrey tests ($p = 0.004$). The coefficient on the 65+ population is $\beta = -2.84$ ($p < 0.001$): each additional million people aged 65+ is associated with a $2.84 billion quarterly reduction in equity mutual fund net flows. The S&P 500 return coefficient is positive but not statistically significant ($\beta = 0.475$, $p = 0.331$). Quarterly equity fund flows from retirement accounts do not track contemporaneous market performance, consistent with the behavioural inertia documented in the retirement savings literature. Appendix Table A6 reports alternative specifications. The coefficient is robust to lag structure (one-quarter lag, $\beta = -2.89$, $p < 0.001$) and the inclusion of a federal funds rate control ($\beta = -2.79$, $p < 0.001$; FFR coefficient $p = 0.857$) (Figure 15). To assess parameter stability, the regression was re-estimated on the pre-pandemic subsample (2007:Q3 to 2019:Q4, N = 50). The demographic coefficient remains significant at the 1% level ($\beta = -2.10$), though attenuated relative to the full-sample estimate of −2.84. This is consistent with the intensification of boomer-cohort outflows after 2020 as larger cohorts crossed the age-65 threshold.

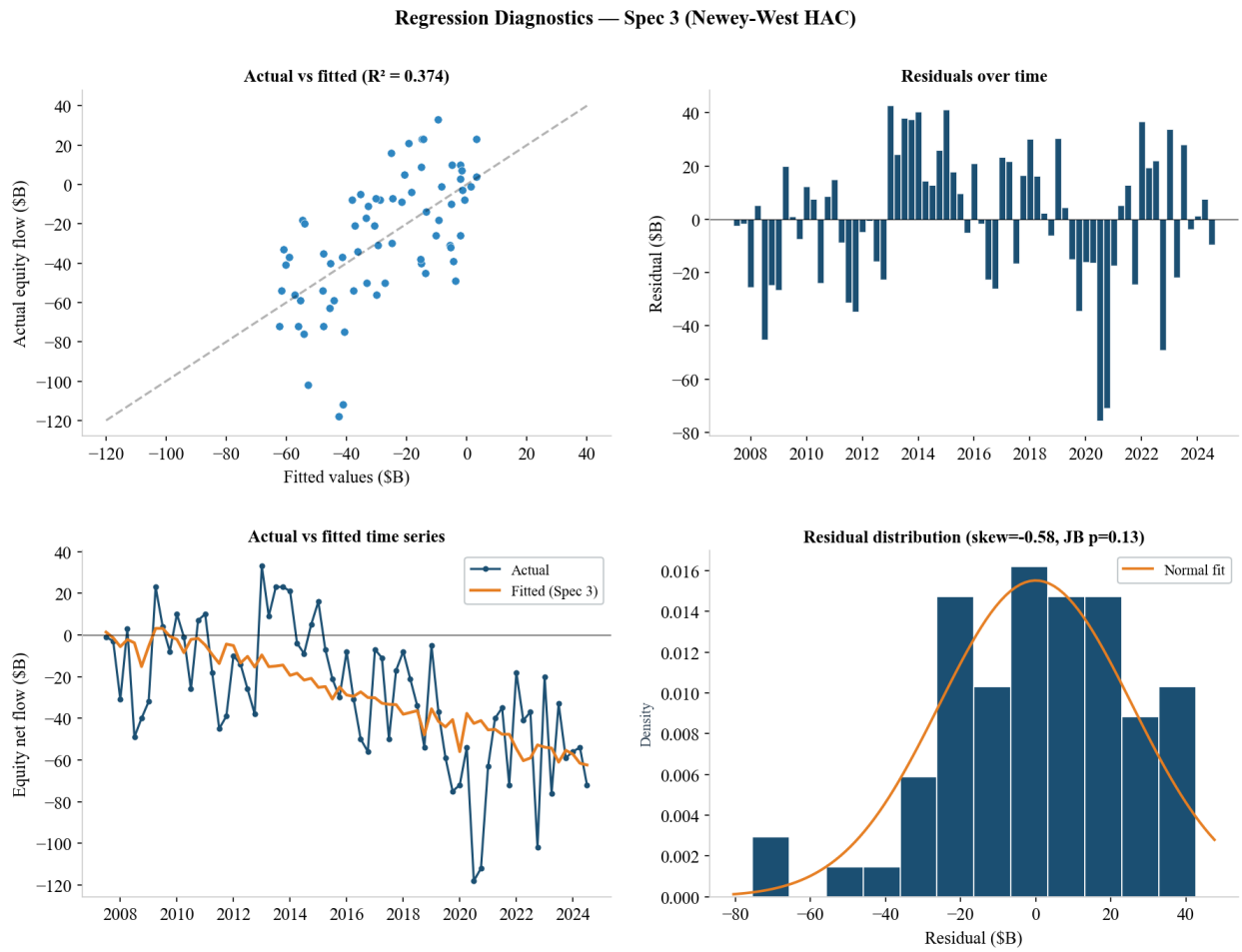

Figure 15: Regression diagnostics. Actual versus fitted, residuals, time-series residuals, and residual distribution.

Table 5

Regression Results: Quarterly Equity Mutual Fund Net Flows on Demographic Structure

Variable	β	SE (NW)	p	R²	N
Population 65+ (millions)	−2.839	0.481	< .001		
S&P 500 quarterly return	0.475	0.487	.331		
Model fit				.374	69

Note. Dependent variable: quarterly net new cash flow into U.S. equity mutual funds ($B), ICI Table 20, 2007:Q3 to 2024:Q3. Newey-West standard errors with four lags. The coefficient on population 65+ indicates that each additional million people aged 65 and older is associated with a $2.84 billion quarterly reduction in equity mutual fund net flows.

To sum up, the structural demographic crossover occurs between 2040 and 2041, with intermittent negative years beginning a decade earlier. Corporate buybacks absorb the crossover in over 95% of simulations, though the offset is procyclical. The annual price drag reaches −0.11% by 2050. Return uncertainty dominates the long-horizon range, but the voluntary withdrawal multiplier is the key structural driver. The regression confirms that the relationship between retirement-age population growth and equity outflows is present in historical data. Section 6 discusses what these findings mean for markets and for policy.

6. Discussion

The structural crossover, the buyback dependency, and the price trajectory each raise questions that the numbers alone cannot answer. This section interprets the findings, tests them against counterarguments, and draws out what they imply.

6.1 The Central Finding

The asset meltdown predicted by the early demographic literature does not materialise under the conservative TDF-based assumptions used here. The median net flow trajectory crosses zero between 2040 and 2041, but the annual price impact at 2050 is −0.11% of market capitalisation, less than one-hundredth of the long-run nominal equity return. Cumulatively, the demographic channel supports prices through 2040 (+$3.5 trillion) before reversing to −$1.3 trillion by 2050. The outcome aligns more closely with the temperate view of Geanakoplos et al. (2004) than with the meltdown thesis. Poterba (2001) concluded that any relationship between demographic structure and equity returns is at most modest, that is consistent with this result. A

drag of −0.11% within a return distribution with 19.5% annual standard deviation would be difficult to detect in the time-series regressions he employed.

Corporate buybacks absorb the structural crossover in 95.5% of simulations, and of the 4.5% that temporarily cross, nearly all revert. Only 0.3% of all simulations show sustained negative total net flows by 2050. The buyback contribution at baseline ($591 billion) is nearly eight times the demographic surplus ($77 billion). The offset is procyclical, contracting 30% to 50% in recessions, but at the projected scale of the demographic drag the buffer is more than sufficient under normal conditions.

The drag is modest in any given year but persistent. It is driven by RMD schedules and population structure, not by market conditions (S&P 500 return coefficient $p = 0.331$), and does not self-correct. This answers the research question posed in Section 1.2. The crossover does occur, but gradually and with modest price consequences, not the meltdown hypothesised in the early demographic literature. The implication is that understanding demographic market pressure requires not better return forecasts but better data on decumulation behaviour. Current knowledge in this domain relies on survey data rather than administrative records.

6.2 Robustness and Sensitivity

Three objections deserve direct engagement, the first concerns multiplier symmetry. The Gabaix-Koijen estimate of $M = 5$ was derived from a sample period (2001 to 2020) dominated by inflows into passive funds. Whether the same multiplier applies to sustained outflows is an open question. Active investors might buy dips more aggressively than they sell rallies, which would imply a lower effective multiplier during outflow periods. The $M = 2$ stress test addresses this. Both the $M = 5$ and $M = 2$ trajectories are essentially zero at 2040 and permanently negative

thereafter, reaching −0.112% and −0.047% by 2050. The multiplier governs scale, not direction. The asymmetry could also run the other way. Large-cap stocks have the highest passive ownership and the lowest demand elasticity (Gabaix & Koijen, 2023; Haddad et al., 2025), potentially making the outflow multiplier higher than the inflow-era estimate.

The second is more subtle. The Sobol analysis shows return dominance at every horizon (ST = 1.064 at 2050), which could be read as evidence that demographics do not matter. But Sobol indices measure what drives output uncertainty, not what drives the output level. The demographic trend shifts net flows downward in every simulation. Return uncertainty determines how large the shift becomes, not whether it exists. Under the deterministic baseline (9.9% geometric mean), the crossover occurs between 2042 and 2043 regardless. The conditional Sobol confirms that, once return uncertainty is removed, crossover timing depends primarily on the voluntary withdrawal multiplier.

A third objection is that markets will equilibrate. If demographic selling depresses prices, value investors will absorb the supply. This general-equilibrium response is real but limited in this context for three reasons. First, the flows modelled here are mechanical. RMDs do not respond to price levels, and TDF contributions do not adjust to expected returns (Giglio et al., 2021, documenting weak belief-to-portfolio pass-through). Second, the equilibrium response operates through the active investor base, but the Haddad et al. (2025) pass-through parameter already captures the degree to which active investors offset flow-driven price displacements (approximately one-third at current passive ownership). Third, the price drag of −0.11% per year is small enough relative to 19.5% annual return volatility that the signal may be indistinguishable from noise for decades. That would delay any corrective response.

6.3 Implications

An annual drag of −0.11% is negligible in isolation, but two features give it practical relevance. First, it concentrates. Demographic outflows through index funds sell in proportion to market-capitalisation weights, and the largest constituents carry the highest passive ownership and lowest demand elasticity (Gabaix & Koijen, 2023; Haddad et al., 2025). The aggregate number understates the pressure on individual large-cap names. Chinco and Sammon (2024) report that standard passive ownership varies by a factor of fifteen between S&P 500 and Russell 2000 constituents, and their correction for quasi-passive strategies applies proportionally across indices. The aggregate model used here does not capture this within-market heterogeneity. Stock-level analysis is a priority extension (Section 6.4). Second, the finding depends entirely on buybacks continuing at historical rates. Any policy or cyclical shock that cuts buyback volumes would expose the underlying demographic imbalance directly. The projected drag is a baseline, not a worst case.

The policy dimension is concrete but limited. A sensitivity test halving the net buyback yield from 0.98% to 0.49% increases the total-flow crossover probability from 4.5% to 5.5%, because even halved buybacks still exceed the demographic deficit through most of the projection horizon. The 1% excise tax on corporate buybacks, enacted in 2022 and proposed for increase in subsequent budget discussions, would need to reduce buyback volumes by substantially more than half before demographic sales become a market-level concern.

For TDF providers, increasing international diversification within glide paths would distribute selling pressure across markets. For policymakers, the systemic role of buybacks warrants consideration before expanding the buyback excise tax. And for long-horizon

institutional investors, return assumptions built on the inflow-dominated era may not fully capture the transition ahead.

6.4 Scope, Limitations, and Future Work

Several forces excluded from the model would increase outflow pressure if incorporated. The SECURE Act's 10-year rule for inherited retirement accounts (effective 2020, with first liquidation deadlines in 2030) creates a second outflow channel as heirs draw down inherited balances. Household debt-driven early withdrawals and the tax incentive to liquidate tax-deferred holdings before taxable ones (step-up in basis) both accelerate equity selling beyond what RMD tables alone predict. U.S. total household debt has risen from approximately $12.4 trillion in 2006 to $18.8 trillion by late 2025 (Bloomberg), with credit card debt alone reaching a record $1.28 trillion. This increases the likelihood that debt-constrained households draw down retirement savings earlier than the model assumes. These omissions mean the projection is a lower bound on total demographic equity outflows.

The model uses mean retirement account balances from the SCF. The conditional mean substantially exceeds the conditional median across all age groups in the SCF (Federal Reserve Board, 2022). The wealthiest households drive a disproportionate share of total withdrawals. The model correctly estimates aggregate outflows but masks heterogeneity across the wealth distribution.

The price impact is not fed back into equity returns within the simulation. The projected drag of approximately 11 basis points per year is small relative to the 19.5% return standard deviation, but over 25 years the cumulative omission could modestly overstate balance growth in

the later projection years. Decomposing historical returns to isolate the DC flow contribution would allow estimation of this feedback bias. It is the most direct extension of the current work.

Parker et al. (2023) document that TDF rebalancing offsets roughly 10% of aggregate trend-chasing flows, which dampens intra-year volatility in both directions. The simulation operates at annual resolution and does not capture this within-year smoothing, which would slightly reduce the variance of net flows without altering the trajectory.

IRA balances are implicitly captured in the outflow estimates because the SCF reports combined retirement account holdings (401(k) and IRA), and RMD rules apply identically to both account types. What the model excludes is a separate IRA inflow channel, since IRA contributions are independent of employer-sponsored DC plans. ICI data confirm that IRA mutual funds exhibit the same structural outflow pattern. IRA domestic equity fund net flows were −$104 billion in 2024 (ICI, 2025). Combined with DC equity fund outflows of −$228 billion, total retirement equity mutual fund outflows reached −$332 billion in 2024. The model's inflow estimate (DC plans only) is a lower bound, while the outflow estimate (SCF-based, capturing both account types) is closer to comprehensive.

ICI data show that DC equity mutual fund flows have been negative since at least 2015, reaching −$228 billion in 2024, while bond mutual fund inflows remained positive at +$48 billion. This cross-asset pattern is consistent with TDF glide-path de-risking rather than wrapper migration, which would produce outflows across all fund types. DC plans cannot generally hold ETFs, so the mutual-fund-to-ETF conversion that contaminates industry-wide flow statistics is minimal for the DC-specific data used here. Some outflows likely reflect migration from mutual fund to collective investment trust wrappers, particularly in hybrid funds (−$58 billion in 2024), but the equity-specific signal appears predominantly demographic. The regression fitted value

for 2024 (−$181 billion for Q1 to Q3) closely matches observed flows. The model does not attempt to decompose these components. Instead, the base case applies TDF equity shares as a forward-looking structural assumption, and the sensitivity test in Section 5.4 quantifies the effect of current retiree allocations.

The simulation stops at 2050, but the demographic mechanism does not. The millennial generation, currently the largest cohort of active 401(k) contributors, will start hitting RMD age in the 2060s and 2070s. That adds a second wave that may be larger in real terms as their balances compound for additional decades.

Future work could go in three directions. First, taxable account flows for a broader measure of retirement equity selling. Second, a stock-level analysis using the Haddad et al. framework to identify where demographic pressure concentrates. Third, the fixed-income side of the TDF balance sheet. At the retirement stage, TDFs allocate roughly 60% to 70% to bonds. In dollar terms, the bond outflow channel is actually larger than the equity channel modelled here. And these sales arrive as governments face elevated borrowing needs with U.S. federal debt-to-GDP above 120%. Demographic bond selling and fiscal issuance pulling in the same direction could produce upward pressure on yields that the equity-only framework does not capture. Spillovers into equity valuations through higher discount rates remain a possibility the model cannot address.

7. Conclusion

This study asked when and whether U.S. retirement account outflows will overtake inflows into passive equity index funds, and what the resulting price impact would be. The answer is yes. There will be net outflows, but gradually. The median net flow trajectory crosses

zero between 2040 and 2041, with intermittent negative years beginning a decade earlier as return volatility causes temporary crossovers. Corporate buybacks absorb the structural crossover in 95.5% of simulations, though the offset is procyclical and contracts in the conditions where demographic pressure is most acute.

The price impact is modest, at −0.11% of market capitalisation per year by 2050, less than one-hundredth of the long-run equity return. The demographic channel supports prices through 2040 before becoming a headwind. The transition plays out over decades, not years. DC equity mutual fund flows have been negative since 2015. The modelled TDF channel remains positive only under the glide-path assumption that understates current retiree equity holdings. The base case represents the structural regime toward which the system is converging, not current conditions.

The main contribution is the five-module pipeline itself. No prior study has combined cohort-level retirement flow accounting with wage-adjusted inflows, the Gabaix-Koijen multiplier, the Haddad et al. time-varying pass-through, and global sensitivity analysis into a single projection framework. The Sobol decomposition reveals that the voluntary withdrawal multiplier is the decisive structural parameter once return uncertainty is removed. The implication is that retirement withdrawal behaviour, not market forecasting, deserves closer attention.

These findings are relevant for TDF providers designing glide paths, for policymakers weighing RMD schedules and buyback taxation, and for institutional investors building long-horizon return assumptions.

The projection is partial-equilibrium and covers only the DC equity channel. Several excluded forces, including inherited account liquidations and household debt-driven withdrawals, would increase outflow pressure if modelled. The clearest next steps involve the fixed-income side of the TDF balance sheet, stock-level passive ownership heterogeneity, and international retirement flows.

References

Abel, A. B. (2001). Will bequests attenuate the predicted meltdown in stock prices when baby boomers retire? The Review of Economics and Statistics, 83(4), 589–595. https://doi.org/10.1162/003465301753237678

Aipinas, A. (2026). capstone-demographic-flows [Source code and data]. GitHub. https://github.com/Aipinas/capstone-demographic-flows

Bloomberg L.P. (2025). Semi-annual passive equity AUM [Data set]. Bloomberg Terminal.

Brooks, R., & Lamont, O. A. (2004). Comments and discussion. Brookings Papers on Economic Activity, 2004(1), 308–325.

Bureau of Labor Statistics. (2024). Employee Benefits in the United States, March 2024. U.S. Department of Labor.

Bureau of Labor Statistics. (2025). Employment Cost Index: Historical listing. U.S. Department of Labor.

Chinco, A., & Sammon, M. (2024). The passive ownership share is double what you think it is. Journal of Financial Economics, 157, 103860. https://doi.org/10.1016/j.jfineco.2024.103860

Damodaran, A. (2025). Historical returns on stocks, bonds and bills: 1928–2024. Stern School of Business, New York University. https://pages.stern.nyu.edu/~adamodar/

FactSet Research Systems. (2025). FT Wilshire 5000 market capitalisation [Data set]. FactSet.

Federal Reserve Board. (2022). Survey of Consumer Finances. Board of Governors of the Federal Reserve System.

Federal Reserve Board. (2025a). Enhanced Financial Accounts: Equity issuance and retirement (Z.1, Table F.103). Board of Governors of the Federal Reserve System.

Gabaix, X., & Koijen, R. S. J. (2023). In search of the origins of financial fluctuations: The inelastic markets hypothesis. NBER Working Paper No. 28967.

Geanakoplos, J., Magill, M., & Quinzii, M. (2004). Demography and the long-run predictability of the stock market. Brookings Papers on Economic Activity, 2004(1), 241–326.

Giglio, S., Maggiori, M., Stroebel, J., & Utkus, S. (2021). Five facts about beliefs and portfolios. American Economic Review, 111(5), 1481–1522. https://doi.org/10.1257/aer.20200243

Haddad, V., Huebner, P., & Loualiche, E. (2025). How competitive is the stock market? Theory, evidence from portfolios, and implications for the rise of passive investing. American Economic Review, 115(3), 975–1018. https://doi.org/10.1257/aer.20230505

Internal Revenue Service. (2024). Publication 590-B: Distributions from Individual Retirement Arrangements (IRAs). U.S. Department of the Treasury.

Investment Company Institute. (2025). 2025 Investment Company Fact Book (65th ed.). Investment Company Institute.

Madrian, B. C., & Shea, D. F. (2001). The power of suggestion: Inertia in 401(k) participation and savings behavior. The Quarterly Journal of Economics, 116(4), 1149–1187. https://doi.org/10.1162/003355301753265543

Mitchell, M., Pedersen, L. H., & Pulvino, T. (2007). Slow moving capital. American Economic Review, 97(2), 215–220. https://doi.org/10.1257/aer.97.2.215

Modigliani, F., & Brumberg, R. (1954). Utility analysis and the consumption function: An interpretation of cross-section data. In K. K. Kurihara (Ed.), Post Keynesian Economics (pp. 388–436). Rutgers University Press.

Morningstar. (2025). 2025 Target-Date Strategy Landscape. Morningstar Research Services.

Parker, J. A., Schoar, A., & Sun, Y. (2023). Retail financial innovation and stock market dynamics: The case of target date funds. The Journal of Finance, 78(5), 2673–2723. https://doi.org/10.1111/jofi.13258

Pension Protection Act of 2006, Pub. L. No. 109–280, 120 Stat. 780.

Poterba, J. M. (2001). Demographic structure and asset returns. The Review of Economics and Statistics, 83(4), 565–584. https://doi.org/10.1162/003465301753237650

Poterba, J. M., Venti, S. F., & Wise, D. A. (2011). The composition and drawdown of wealth in retirement. Journal of Economic Perspectives, 25(4), 95–118. https://doi.org/10.1257/jep.25.4.95

S&P Dow Jones Indices. (2024). S&P 500 Buyback Quarterly Reports [Data set]. S&P Global.

Sabbatucci, R., Tamoni, A., & Xiao, S. (2025). Shifting from active to passive: How retirement plans impact equity prices. Swedish House of Finance Research Paper No. 23–08.

SECURE 2.0 Act of 2022, Pub. L. No. 117–328, Div. T, 136 Stat. 4459.

U.S. Census Bureau. (2023). 2023 National Population Projections. U.S. Department of Commerce.

Vanguard. (2025a). How America Retires 2025. The Vanguard Group.

Vanguard. (2025b). How America Saves 2025. The Vanguard Group.

Appendices

Appendix A: Data Sources and Parameter Tables

The tables below report the calibrated parameters for each module. All processing code and raw data are available in the project's GitHub repository (github.com/Aipinas/capstone-demographic-flows).

Table A1

Module 2: Inflow Parameters

Age Group	Salary ($)	Partic.	Contrib.	TDF US Eq.	Emp. Ratio	$/Worker
<25	$39,979	54.0%	9.3%	53.6%	28.2%	$1,076
25–34	$59,800	82.0%	11.1%	53.5%	65.3%	$2,912
35–44	$72,020	86.0%	11.7%	53.0%	66.4%	$3,841
45–54	$71,604	87.0%	12.3%	44.4%	63.2%	$3,402
55–64	$68,744	87.0%	13.8%	35.6%	48.6%	$2,938
65+	$62,036	79.0%	14.6%	18.5%	9.3%	$1,324

Note. Sources: BLS National Compensation Survey (2024), Vanguard How America Saves (2025b), Vanguard TDF prospectuses. Plan access rate: 72% (BLS NCS). Nominal wage growth: 2.84%/year (BLS Employment Cost Index, 2001–2025).

Table A2

Module 3: Outflow Parameters

Age Group	Balance ($K)	Own. %	Eff. Rate	TDF Eq. %	RMD Rate	Per-Cap Eq ($K)
55–59	537.6	57.0%	1.00%	35.6%	—	1.09
60–64	537.6	57.0%	2.00%	30.0%	—	1.84
65–72	609.2	51.0%	3.00%	21.0%	—	1.96
73–74	609.2	51.0%	3.85%	19.5%	3.85%	2.33
75–84	462.4	42.0%	4.95%	18.0%	4.95%	1.73
85–94	462.4	42.0%	8.20%	18.0%	8.20%	2.87
95+	462.4	42.0%	12.82%	18.0%	12.82%	4.48

Note. Sources: Federal Reserve SCF (2022), IRS Publication 590-B Table III, Vanguard How America Retires (2025a), Vanguard TDF prospectuses. Effective rate = max(RMD rate, voluntary rate). Voluntary rates: 1% (55–59), 2% (60–64), 3% (65–72); above 73 RMD binds. Ownership convergence: 2.7%/year. Dashes indicate voluntary rate exceeds RMD.

Table A3

Module 4: Buyback Calibration

Parameter	Value
Net buyback yield (mean)	0.98% of market capitalisation
Net buyback yield (std dev)	0.72%
Source	Federal Reserve Z.1 Table F.103, 2020–2024
Procyclical cut (return < −10%)	−30% (mild recession)
Procyclical cut (return < −20%)	−50% (severe recession)
Cross-validation	S&P Dow Jones Quarterly Buyback Reports

Note. Calibrated from Federal Reserve Z.1 Enhanced Financial Accounts (Table F.103), 2020–2024.

Table A4

Module 5: Price Impact Calibration

Parameter	Value
Base multiplier M	U(3, 7); central estimate 5 (Gabaix & Koijen, 2023)
Passive share, 2025 baseline	35% (Bloomberg, scaled ×1.583 per Chinco & Sammon, 2024)
ARIMA(0,1,0) drift	+0.46 pp per half-year
Passive share cap	67%
Haddad et al. strategic response χ	3
Pass-through at 2025	0.34
Pass-through at 2050	0.50
Effective multiplier at 2050	≈6.0

Note. Pass-through from Haddad et al. (2025): $1 / (1 + \chi \times \text{active share})$. Passive share scaled by 1.583 to align with Chinco & Sammon (2024).

Table A5

ARIMA Model Comparison

Model	AIC	BIC	Log-L
ARIMA(0,1,0) with drift	42.9	45.4	−19.5
ARIMA(1,1,0) with drift	43.3	47.0	−18.6
ARIMA(0,1,1) with drift	43.6	47.4	−18.8

Note. Referenced in §4.3. Fitted to 27 semi-annual observations. Best AIC: ARIMA(0,1,0) with drift +0.46 pp/half-year. RMSE: 1.93 pp.

Table A6

Regression Robustness

Specification	β pop65+	SE	p	R^2	N
(a) Base	−2.839	0.481	< .001	.374	69
(b) Lagged pop	−2.892	0.505	< .001	.373	68
(c) FFR control	−2.793	0.645	< .001	.374	69

Note. Referenced in §5.6. Newey-West SE with four lags. FFR coefficient in spec (c): $\beta = -0.422$, $p = .857$.

Table A7

Immigration Scenario Disaggregation

Scenario	Crossover Probability	Median Crossover Year
Mid	90.6%	2031
High immigration	89.1%	2031
Low immigration	91.5%	2031

Note. Referenced in §4.1 and §5.1. Spread: 2.4 pp. High immigration produces lowest crossover probability (larger working-age population generates higher inflows).

GitHub repository: github.com/Aipinas/capstone-demographic-flows

Appendix Figures

Figure A-1: Monte Carlo Convergence

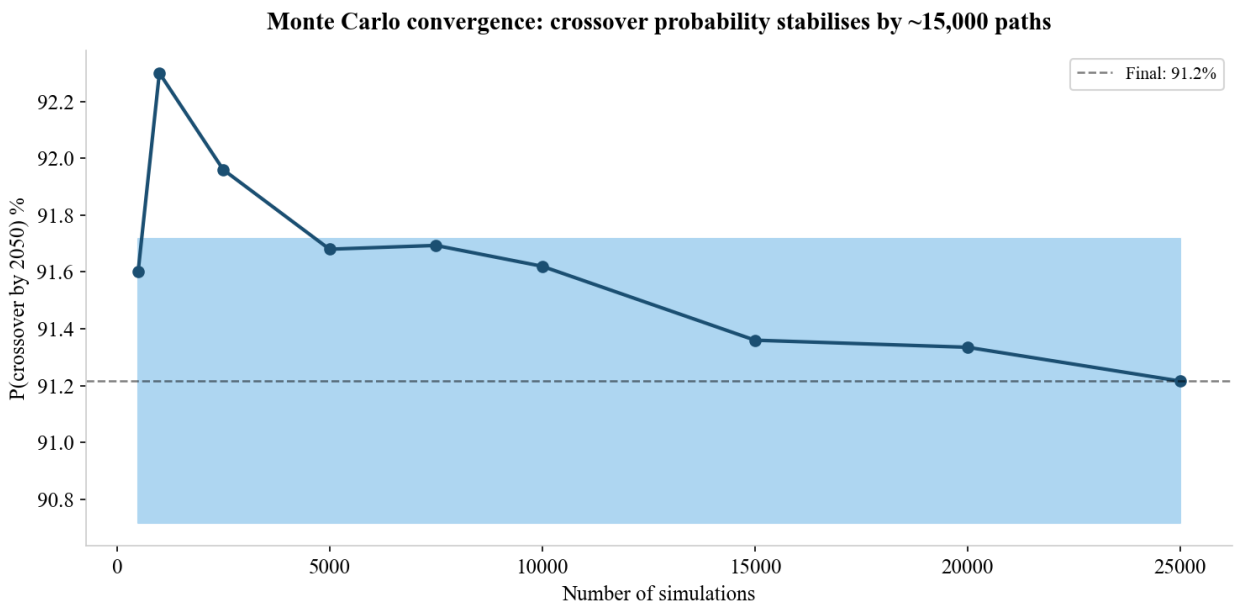
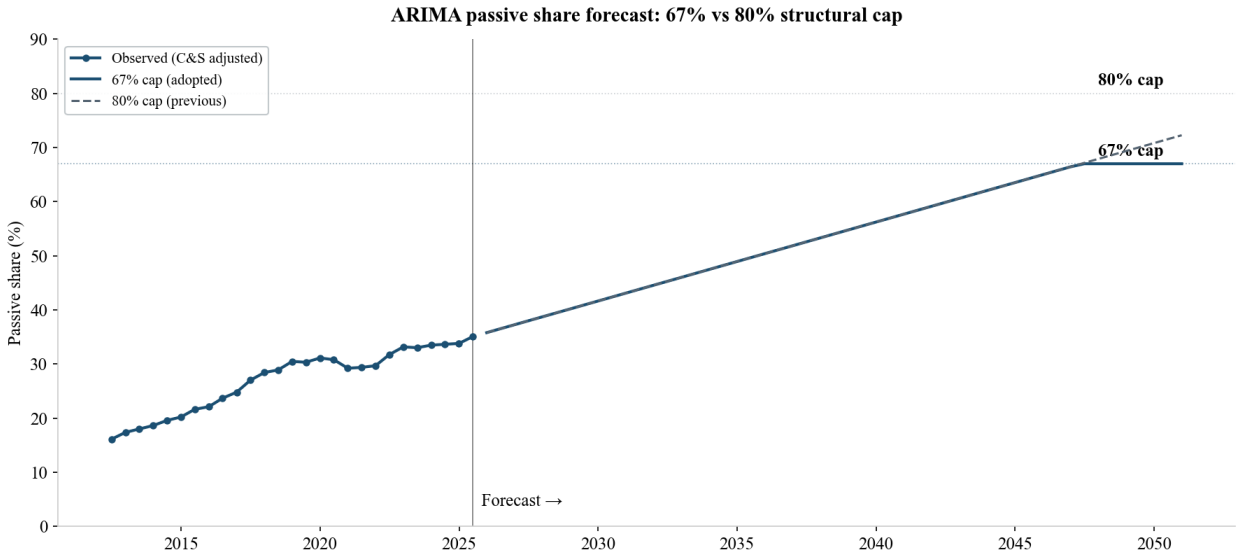
Figure A-2: Passive Share Cap Comparison (67% vs 80%)